

QUESTION: 1

A company employs a team of customer service agents to provide telephone and email support to customers.

The company develops a webchat bot to provide automated answers to common customer queries. Which business benefit should the company expect as a result of creating the webchat bot solution?

- A. increased sales
- B. a reduced workload for the customer service agents
- C. improved product reliability

Answer(s): B

QUESTION: 2

For a machine learning progress, how should you split data for training and evaluation?

- A. Use features for training and labels for evaluation.
- B. Randomly split the data into rows for training and rows for evaluation.
- C. Use labels for training and features for evaluation.
- D. Randomly split the data into columns for training and columns for evaluation.

Answer(s): B

Explanation:

The Split Data module is particularly useful when you need to separate data into training and testing sets. Use the Split Rows option if you want to divide the data into two parts. You can specify the percentage of data to put in each split, but by default, the data is divided 50-50. You can also randomize the selection of rows in each group, and use stratified sampling.

Reference:

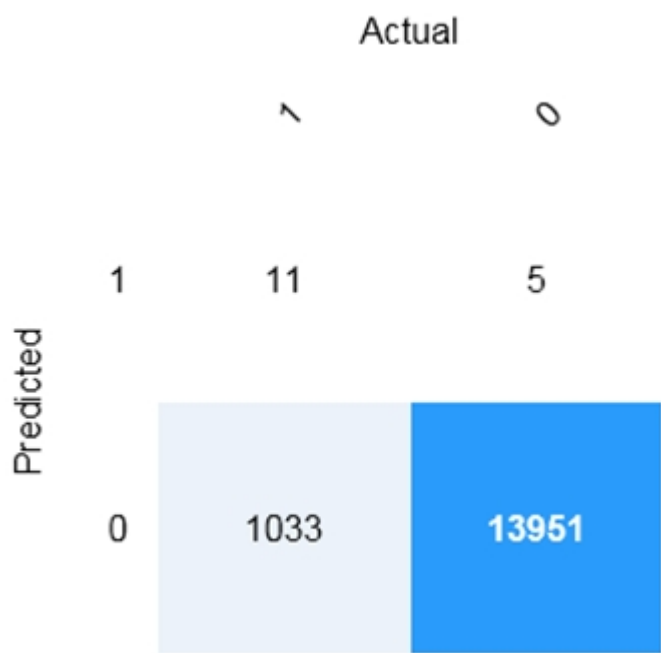
<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/split-data>

QUESTION: 3

HOTSPOT (Drag and Drop is not supported)

You are developing a model to predict events by using classification.

You have a confusion matrix for the model scored on test data as shown in the following exhibit.



Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

There are [answer choice] correctly predicted positives.

▼

5
11
1,033
13,951

There are [answer choice] false negatives.

▼

5
11
1,033
13,951

Answer(s): A

Explanation:

Answer Area

There are [answer choice] correctly predicted positives.

5

11

1,033

13,951

There are [answer choice] false negatives.

5

11

1,033

13,951

Box 1: 11

	Predicted	
	Positive	Negative
Actual True	TP	FN
Actual False	FP	TN

TP = True Positive.

The class labels in the training set can take on only two possible values, which we usually refer to as positive or negative. The positive and negative instances that a classifier predicts correctly are called true positives (TP) and true negatives (TN), respectively. Similarly, the incorrectly classified instances are called false positives (FP) and false negatives (FN).

Box 2: 1,033

FN = False Negative

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio/evaluate-model-performance>

QUESTION: 4

You build a machine learning model by using the automated machine learning user interface (UI). You need to ensure that the model meets the Microsoft transparency principle for responsible AI. What should you do?

- A. Set Validation type to Auto.
- B. Enable Explain best model.

- C. Set Primary metric to accuracy.
- D. Set Max concurrent iterations to 0.

Answer(s): B

Explanation:

Model Explain Ability.

Most businesses run on trust and being able to open the ML “black box” helps build transparency and trust. In heavily regulated industries like healthcare and banking, it is critical to comply with regulations and best practices. One key aspect of this is understanding the relationship between input variables (features) and model output. Knowing both the magnitude and direction of the impact each feature (feature importance) has on the predicted value helps better understand and explain the model. With model explain ability, we enable you to understand feature importance as part of automated ML runs.

Reference:

<https://azure.microsoft.com/en-us/blog/new-automated-machine-learning-capabilities-in-azure-machine-learning-service/>

QUESTION: 5

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Forecasting housing prices based on historical data is an example of anomaly detection.	☐	☐
Identifying suspicious sign-ins by looking for deviations from usual patterns is an example of anomaly detection.	☐	☐
Predicting whether a patient will develop diabetes based on the patient's medical history is an example of anomaly detection.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Forecasting housing prices based on historical data is an example of anomaly detection.	☐	☒
Identifying suspicious sign-ins by looking for deviations from usual patterns is an example of anomaly detection.	☒	☐
Predicting whether a patient will develop diabetes based on the patient's medical history is an example of anomaly detection.	☐	☒

Anomaly detection encompasses many important tasks in machine learning:

Identifying transactions that are potentially fraudulent.

Learning patterns that indicate that a network intrusion has occurred.

Finding abnormal clusters of patients.

Checking values entered into a system.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/anomaly-detection>

QUESTION: 6

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

The handling of unusual or missing values provided to an AI system is a consideration for the Microsoft principle for responsible AI.

- inclusiveness
- privacy and security
- reliability and safety
- transparency

Answer(s): A

Explanation:

Answer Area

The handling of unusual or missing values provided to an AI system is a consideration for the Microsoft principle for responsible AI.

- inclusiveness
- privacy and security
- reliability and safety
- transparency

Reliability and safety:

AI systems need to be reliable and safe in order to be trusted. It is important for a system to perform as it was originally designed and for it to respond safely to new situations. Its inherent resilience should resist intended or unintended manipulation. Rigorous testing and validation should be established for operating conditions to ensure that the system responds safely to edge cases, and A/B testing and champion/challenger methods should be integrated into the evaluation process.

An AI system's performance can degrade over time, so a robust monitoring and model tracking process needs to be established to reactively and proactively measure the model's performance and retrain it, as necessary, to modernize it.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

QUESTION: 7

DRAG DROP (Drag and Drop is not supported)

Match the types of AI workloads to the appropriate scenarios.

To answer, drag the appropriate workload type from the column on the left to its scenario on the right. Each workload type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Workloads Types

Anomaly detection

Computer vision

Conversational AI

Knowledge mining

Natural language processing

Answer Area

Workload Type

An automated chat to answer questions about refunds and exchange

Workload Type

Determining whether a photo contains a person

Workload Type

Determining whether a review is positive or negative

Answer(s): A

Explanation:

Workloads Types

Anomaly detection

Computer vision

Conversational AI

Knowledge mining

Natural language processing

Answer Area

Conversational AI

An automated chat to answer questions about refunds and exchange

Computer vision

Determining whether a photo contains a person

Natural language processing

Determining whether a review is positive or negative

Box 3: Natural language processing

Natural language processing (NLP) is used for tasks such as sentiment analysis, topic detection, language detection, key phrase extraction, and document categorization.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 8

You are designing an AI system that empowers everyone, including people who have hearing, visual, and other impairments.

This is an example of which Microsoft guiding principle for responsible AI?

- A. fairness
- B. inclusiveness
- C. reliability and safety
- D. accountability

Answer(s): B

Explanation:

Inclusiveness: At Microsoft, we firmly believe everyone should benefit from intelligent technology, meaning it must incorporate and address a broad range of human needs and experiences. For the 1 billion people with disabilities around the world, AI technologies can be a game-changer.

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 9

DRAG DROP (Drag and Drop is not supported)

Match the Microsoft guiding principles for responsible AI to the appropriate descriptions.

To answer, drag the appropriate principle from the column on the left to its description on the right. Each principle may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Principles

Accountability

Fairness

Inclusiveness

Privacy and security

Reliability and safety

Answer Area

Principle

Ensure that AI systems operate as they were originally designed, respond to unanticipated conditions, and resist harmful manipulation.

Principle

Implementing processes to ensure that decisions made by AI systems can be overridden by humans.

Principle

Provide consumers with information and controls over the collection, use, and storage of their data.

Answer(s): A

Explanation:

Principles

Accountability

Fairness

Inclusiveness

Privacy and security

Reliability and safety

Answer Area

Reliability and safety

Ensure that AI systems operate as they were originally designed, respond to unanticipated conditions, and resist harmful manipulation.

Accountability

Implementing processes to ensure that decisions made by AI systems can be overridden by humans.

Privacy and security

Provide consumers with information and controls over the collection, use, and storage of their data.

Box 1: Reliability and safety

To build trust, it's critical that AI systems operate reliably, safely, and consistently under normal circumstances and in unexpected conditions. These systems should be able to operate as they were originally designed, respond safely to unanticipated conditions, and resist harmful manipulation.

Box 2: Accountability

The people who design and deploy AI systems must be accountable for how their systems operate. Organizations should draw upon industry standards to develop accountability norms. These norms can ensure that AI systems are not the final authority on any decision that impacts people's lives and that humans maintain meaningful control over otherwise highly autonomous AI systems.

Box 3: Privacy and security

As AI becomes more prevalent, protecting privacy and securing important personal and business

information is becoming more critical and complex. With AI, privacy and data security issues require especially close attention because access to data is essential for AI systems to make accurate and informed predictions and decisions about people. AI systems must comply with privacy laws that require transparency about the collection, use, and storage of data and mandate that consumers have appropriate controls to choose how their data is used

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 10

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

When developing an AI system for self-driving cars, the Microsoft for responsible AI should be applied to ensure consistent operation system during unexpected circumstances.

▼
inclusiveness
accountability
reliability and safety
fairness

principle
of the

Answer(s): A

Explanation:

When developing an AI system for self-driving cars, the Microsoft for responsible AI should be applied to ensure consistent operation system during unexpected circumstances.

▼
inclusiveness
accountability
reliability and safety
fairness

principle
of the

Reliability and safety: To build trust, it's critical that AI systems operate reliably, safely, and consistently under normal circumstances and in unexpected conditions.

These systems should be able to operate as they were originally designed, respond safely to unanticipated conditions, and resist harmful manipulation.

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 11

You are building an AI system.

Which task should you include to ensure that the service meets the Microsoft transparency principle for responsible AI?

- A. Ensure that all visuals have an associated text that can be read by a screen reader.
- B. Enable autoscaling to ensure that a service scales based on demand.
- C. Provide documentation to help developers debug code.
- D. Ensure that a training dataset is representative of the population.

Answer(s): C

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 12

DRAG DROP (Drag and Drop is not supported)

Match the types of AI workloads to the appropriate scenarios.

To answer, drag the appropriate workload type from the column on the left to its scenario on the right. Each workload type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Workload Types

Anomaly detection

Computer vision

Machine Learning (Regression)

Natural language processing

Answer Area

Workload Type

Identify handwritten letters.

Workload Type

Predict the sentiment of a social media post.

Workload Type

Identify a fraudulent credit card payment.

Workload Type

Predict next month's toy sales.

Answer(s): A

Explanation:

Workload Types

Anomaly detection

Computer vision

Machine Learning (Regression)

Natural language processing

Answer Area

Computer vision

Identify handwritten letters.

Natural language processing

Predict the sentiment of a social media post.

Anomaly detection

Identify a fraudulent credit card payment.

Machine Learning (Regression)

Predict next month's toy sales.

Reference:

<https://docs.microsoft.com/en-us/learn/paths/get-started-with-artificial-intelligence-on-azure/>

QUESTION: 13

Your company is exploring the use of voice recognition technologies in its smart home devices. The company wants to identify any barriers that might unintentionally leave out specific user groups. This is an example of which Microsoft guiding principle for responsible AI?

- A. accountability
- B. fairness
- C. inclusiveness
- D. privacy and security

Answer(s): C

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 14

What are three Microsoft guiding principles for responsible AI? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. knowledgeability
- B. decisiveness
- C. inclusiveness
- D. fairness
- E. opinionatedness
- F. reliability and safety

Answer(s): C,D,F

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 15

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Returning a bounding box that indicates the location of a vehicle in an image is an example of

image classification.
object detection.
optical character recognizer (OCR).
semantic segmentation.

Answer(s): A

Explanation:

Answer Area

Returning a bounding box that indicates the location of a vehicle in an image is an example of

image classification.
object detection.
optical character recognizer (OCR).
semantic segmentation.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-object-detection>

QUESTION: 16

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Feature engineering
Feature selection
Model evaluation
Model training

is used to generate additional features.

Answer(s): A

Explanation:

Answer Area

	▼
Feature engineering	
Feature selection	
Model evaluation	
Model training	

is used to generate additional features.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/team-data-science-process/create-features>

QUESTION: 17

You run a charity event that involves posting photos of people wearing sunglasses on Twitter. You need to ensure that you only retweet photos that meet the following requirements:

- Include one or more faces.
- Contain at least one person wearing sunglasses.

What should you use to analyze the images?

- A. the Verify operation in the Face service
- B. the Detect operation in the Face service
- C. the Describe Image operation in the Computer Vision service
- D. the Analyze Image operation in the Computer Vision service

Answer(s): B

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/face/overview>

QUESTION: 18

When you design an AI system to assess whether loans should be approved, the factors used to make the decision should be explainable.

This is an example of which Microsoft guiding principle for responsible AI?

- A. transparency
- B. inclusiveness
- C. fairness
- D. privacy and security

Answer(s): A

Explanation:

Achieving transparency helps the team to understand the data and algorithms used to train the model, what transformation logic was applied to the data, the final model generated, and its associated assets. This information offers insights about how the model was created, which allows it to be reproduced in a transparent way.

Incorrect Answers:

B: Inclusiveness mandates that AI should consider all human races and experiences, and inclusive design practices can help developers to understand and address potential barriers that could unintentionally exclude people.

Where possible, speech-to-text, text-to-speech, and visual recognition technology should be used to empower people with hearing, visual, and other impairments.

C: Fairness is a core ethical principle that all humans aim to understand and apply. This principle is even more important when AI systems are being developed.

Key checks and balances need to make sure that the system's decisions don't discriminate or run a gender, race, sexual orientation, or religion bias toward a group or individual.

D: A data holder is obligated to protect the data in an AI system, and privacy and security are an integral part of this system. Personal needs to be secured, and it should be accessed in a way that doesn't compromise an individual's privacy.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/strategy/responsible-ai>

QUESTION: 19

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Providing an explanation of the outcome of a credit loan application is an example of the Microsoft transparency principle for responsible AI.	☐	☐
A triage bot that prioritizes insurance claims based on injuries is an example of the Microsoft reliability and safety principle for responsible AI.	☐	☐
An AI solution that is offered at different prices for different sales territories is an example of the Microsoft inclusiveness principle for responsible AI.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Providing an explanation of the outcome of a credit loan application is an example of the Microsoft transparency principle for responsible AI.	☒	☐
A triage bot that prioritizes insurance claims based on injuries is an example of the Microsoft reliability and safety principle for responsible AI.	☐	☒
An AI solution that is offered at different prices for different sales territories is an example of the Microsoft inclusiveness principle for responsible AI.	☐	☒

Box 1: Yes

Achieving transparency helps the team to understand the data and algorithms used to train the model, what transformation logic was applied to the data, the final model generated, and its associated assets. This information offers insights about how the model was created, which allows it to be reproduced in a transparent way.

Box 2: No

A data holder is obligated to protect the data in an AI system, and privacy and security are an integral part of this system. Personal needs to be secured, and it should be accessed in a way that doesn't compromise an individual's privacy.

Box 3: No

Inclusiveness mandates that AI should consider all human races and experiences, and inclusive design practices can help developers to understand and address potential barriers that could unintentionally exclude people.

Where possible, speech-to-text, text-to-speech, and visual recognition technology should be used to empower people with hearing, visual, and other impairments.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

QUESTION: 20

DRAG DROP (Drag and Drop is not supported)

Match the principles of responsible AI to appropriate requirements.

To answer, drag the appropriate principles from the column on the left to its requirement on the right. Each principle may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Principles

- Fairness
- Privacy and security
- Reliability and safety
- Transparency

Answer Area

- The system must not discriminate based on gender, race
- Personal data must be visible only to approve
- Automated decision-making processes must be recorded so that approved users can identify why a decision was made

Answer(s): A

Explanation:

Principles

- Fairness
- Privacy and security
- Reliability and safety
- Transparency

Answer Area

- Fairness
- Privacy and security
- Transparency

- The system must not discriminate based on gender, race
- Personal data must be visible only to approve
- Automated decision-making processes must be recorded so that approved users can identify why a decision was made

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>
<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

QUESTION: 21

DRAG DROP (Drag and Drop is not supported)

You plan to deploy an Azure Machine Learning model as a service that will be used by client applications. Which three processes should you perform in sequence before you deploy the model? To answer, move the appropriate processes from the list of processes to the answer area and arrange them in the correct order. Select and Place:

Processes

data encryption

model retraining

model training

data preparation

model evaluation



Answer Area

Answer(s): A

Explanation:

Processes

data encryption

model retraining



Answer Area

data preparation

model training

model evaluation

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-ml-pipelines>

QUESTION: 22

You are building an AI-based app.

You need to ensure that the app uses the principles for responsible AI.

Which two principles should you follow? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. Implement an Agile software development methodology
- B. Implement a process of AI model validation as part of the software review process
- C. Establish a risk governance committee that includes members of the legal team, members of the risk management team, and a privacy officer
- D. Prevent the disclosure of the use of AI-based algorithms for automated decision making

Answer(s): B,C

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/3-implications-responsible-ai-practical>

QUESTION: 23

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

According to Microsoft's

	▼
accountability	
fairness	
inclusiveness	
transparency	

principle of responsible AI,

AI systems should **NOT** reflect biases from the data sets that are used to train the systems.

Answer(s): A

Explanation:

Answer Area

According to Microsoft's

	▼
accountability	
fairness	
inclusiveness	
transparency	

principle of responsible AI,

AI systems should **NOT** reflect biases from the data sets that are used to train the systems.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

QUESTION: 24

HOTSPOT (Drag and Drop is not supported)

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

According to Microsoft's

	▼
accountability	
fairness	
inclusiveness	
transparency	

principle of responsible AI,

AI systems should **NOT** reflect biases from the data sets that are used to train the systems.

Answer(s): A

Explanation:

Answer Area

According to Microsoft's

	▼
accountability	
fairness	
inclusiveness	
transparency	

principle of responsible AI,

AI systems should **NOT** reflect biases from the data sets that are used to train the systems.

Fairness is a core ethical principle that all humans aim to understand and apply. This principle is even more important when AI systems are being developed. Key checks and balances need to make sure that the system's decisions don't discriminate or run a gender, race, sexual orientation, or religion bias toward a group or individual.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

QUESTION: 25

DRAG DROP (Drag and Drop is not supported)

Match the types of AI workloads to the appropriate scenarios.

To answer, drag the appropriate workload type from the column on the left to its scenario on the right. Each workload type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Workload Types

Anomaly detection

Computer vision

Knowledge mining

Natural language processing

Answer Area

Workload type

An automated chatbot to answer questions about refunds and exchanges

Workload type

Determining whether a photo contains a person

Workload type

Determining whether a review is positive or negative

Answer(s): A

Explanation:

Workload Types

Anomaly detection

Computer vision

Knowledge mining

Natural language processing

Answer Area

Knowledge mining

An automated chatbot to answer questions about refunds and exchanges

Computer vision

Determining whether a photo contains a person

Natural language processing

Determining whether a review is positive or negative

Box 1: Knowledge mining

You can use Azure Cognitive Search's knowledge mining results and populate your knowledge base of your

chatbot.

Box 2: Computer vision

Box 3: Natural language processing

Natural language processing (NLP) is used for tasks such as sentiment analysis.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 26

DRAG DROP (Drag and Drop is not supported)

Match the machine learning tasks to the appropriate scenarios.

To answer, drag the appropriate task from the column on the left to its scenario on the right. Each task may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Learning Types

Feature engineering

Feature selection

Model deployment

Model evaluation

Model training

Answer Area

Task

Examining the values of a confusion matrix

Task

Splitting a date into month, day, and year fields

Task

Picking temperature and pressure to train a weather model

Answer(s): A

Explanation:

Learning Types

Feature engineering

Feature selection

Model deployment

Model evaluation

Model training

Answer Area

Model evaluation

Examining the values of a confusion matrix

Feature engineering

Splitting a date into month, day, and year fields

Feature selection

Picking temperature and pressure to train a weather model

Box 1: Model evaluation

The Model evaluation module outputs a confusion matrix showing the number of true positives, false

negatives, false positives, and true negatives, as well as ROC, Precision/Recall, and Lift curves.

Box 2: Feature engineering

Feature engineering is the process of using domain knowledge of the data to create features that help ML algorithms learn better. In Azure Machine Learning, scaling and normalization techniques are applied to facilitate feature engineering. Collectively, these techniques and feature engineering are referred to as featurization.

Note: Often, features are created from raw data through a process of feature engineering. For example, a time stamp in itself might not be useful for modeling until the information is transformed into units of days, months, or categories that are relevant to the problem, such as holiday versus working day.

Box 3: Feature selection

In machine learning and statistics, feature selection is the process of selecting a subset of relevant, useful features to use in building an analytical model. Feature selection helps narrow the field of data to the most valuable inputs. Narrowing the field of data helps reduce noise and improve training performance.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio/evaluate-model-performance>

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-automated-ml>

QUESTION: 27

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Data values that influence the prediction of a model are called

	▼
dependant variables.	
features.	
identifiers.	
labels.	

Answer(s): A

Explanation:

Answer Area

Data values that influence the prediction of a model are called

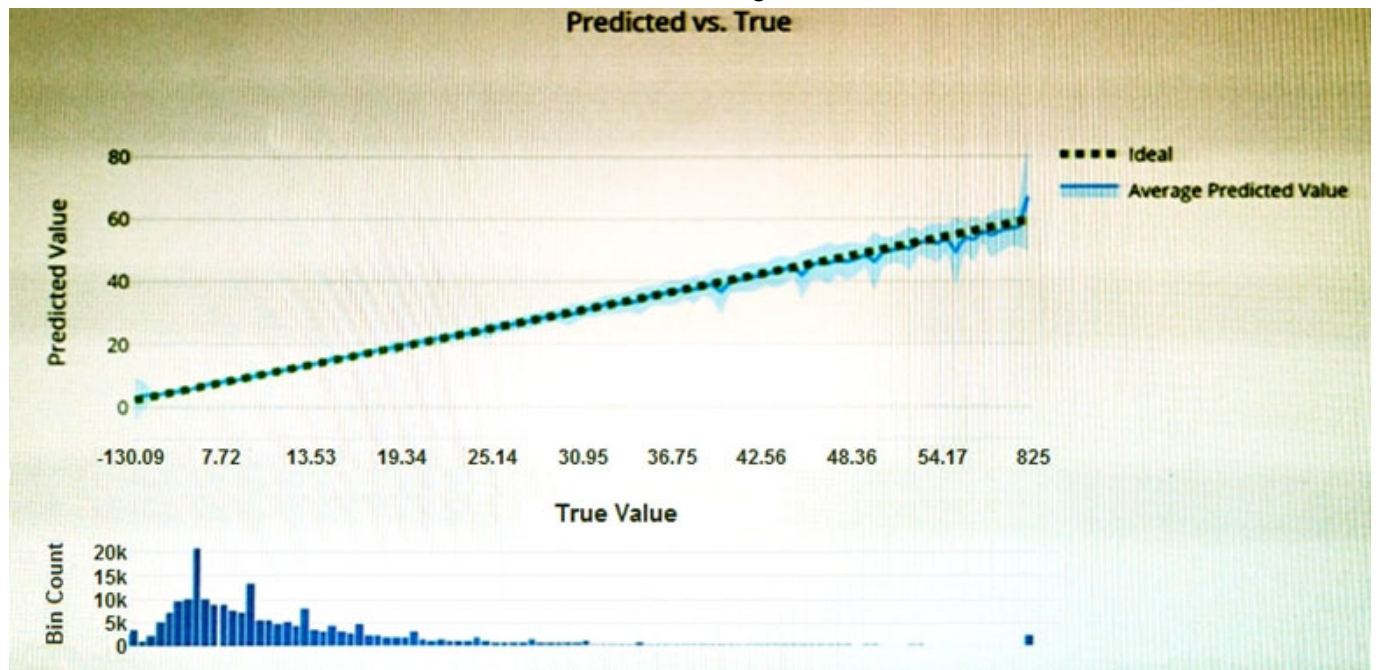
dependant variables.
features.
identifiers.
labels.

Reference:

<https://www.baeldung.com/cs/feature-vs-label> <https://machinelearningmastery.com/discover-feature-engineering-how-to-engineer-features-and-how-to-get-good-at-it/>

QUESTION: 28

You have the Predicted vs. True chart shown in the following exhibit.



Which type of model is the chart used to evaluate?

- A. classification
- B. regression
- C. clustering

Answer(s): B

Explanation:

What is a Predicted vs. True chart?

Predicted vs. True shows the relationship between a predicted value and its correlating true value for a

regression problem. This graph can be used to measure performance of a model as the closer to the $y=x$ line the predicted values are, the better the accuracy of a predictive model.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/how-to-understand-automated-m>

QUESTION: 29

Which type of machine learning should you use to predict the number of gift cards that will be sold next month?

- A. classification
- B. regression
- C. clustering

Answer(s): B

Explanation:

In the most basic sense, regression refers to prediction of a numeric target.

Linear regression attempts to establish a linear relationship between one or more independent variables and a numeric outcome, or dependent variable.

You use this module to define a linear regression method, and then train a model using a labeled dataset. The trained model can then be used to make predictions.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/linear-regression>

QUESTION: 30

You have a dataset that contains information about taxi journeys that occurred during a given period.

You need to train a model to predict the fare of a taxi journey.

What should you use as a feature?

- A. the number of taxi journeys in the dataset
- B. the trip distance of individual taxi journeys
- C. the fare of individual taxi journeys
- D. the trip ID of individual taxi journeys

Answer(s): B

Explanation:

The label is the column you want to predict. The identified Features are the inputs you give the model to predict the Label.

Example:

The provided data set contains the following columns:

vendor_id: The ID of the taxi vendor is a feature.

rate_code: The rate type of the taxi trip is a feature.

passenger_count: The number of passengers on the trip is a feature. trip_time_in_secs: The amount of time the trip took. You want to predict the fare of the trip before the trip is completed. At that moment, you don't know how long the trip would take. Thus, the trip time is not a feature and you'll exclude this column from the model. trip_distance: The distance of the trip is a feature. payment_type: The payment method (cash or credit card) is a feature. fare_amount: The total taxi fare paid is the label.

Reference:

<https://docs.microsoft.com/en-us/dotnet/machine-learning/tutorials/predict-prices>

QUESTION: 31

You need to predict the sea level in meters for the next 10 years.

Which type of machine learning should you use?

- A. classification
- B. regression
- C. clustering

Answer(s): B

Explanation:

In the most basic sense, regression refers to prediction of a numeric target.

Linear regression attempts to establish a linear relationship between one or more independent variables and a numeric outcome, or dependent variable.

You use this module to define a linear regression method, and then train a model using a labeled dataset.

The trained model can then be used to make predictions.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/linear-regression>

QUESTION: 32

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Automated machine learning is the process of automating the time-consuming, iterative tasks of machine learning model development.	☐	☐
Automated machine learning can automatically infer the training data from the use case provided.	☐	☐
Automated machine learning works by running multiple training iterations that are scored and ranked by the metrics you specify.	☐	☐
Automated machine learning enables you to specify a dataset and will automatically understand which label to predict.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Automated machine learning is the process of automating the time-consuming, iterative tasks of machine learning model development.	☒	☐
Automated machine learning can automatically infer the training data from the use case provided.	☐	☒
Automated machine learning works by running multiple training iterations that are scored and ranked by the metrics you specify.	☒	☐
Automated machine learning enables you to specify a dataset and will automatically understand which label to predict.	☐	☒

Box 1: Yes

Automated machine learning, also referred to as automated ML or AutoML, is the process of automating the time consuming, iterative tasks of machine learning model development. It allows data scientists, analysts, and developers to build ML models with high scale, efficiency, and productivity all while sustaining model quality.

Box 2: No

Box 3: Yes

During training, Azure Machine Learning creates a number of pipelines in parallel that try different algorithms and parameters for you. The service iterates through

ML algorithms paired with feature selections, where each iteration produces a model with a training score. The higher the score, the better the model is considered to "fit" your data. It will stop once it hits the exit criteria defined in the experiment.

Box 4: No

Apply automated ML when you want Azure Machine Learning to train and tune a model for you using the target metric you specify.

The label is the column you want to predict.

Reference:

<https://azure.microsoft.com/en-us/services/machine-learning/automatedml/#features>

QUESTION: 33

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

A banking system that predicts whether a loan will be repaid is an example of the type of machine learning.

classification

regression

clustering

Answer(s): A

Explanation:

Answer Area

A banking system that predicts whether a loan will be repaid is an example of the type of machine learning.

classification

regression

clustering

Two-class classification provides the answer to simple two-choice questions such as Yes/No or True/False.

QUESTION: 34

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Labelling is the process of tagging training data with known values.	☐	☐
You should evaluate a model by using the same data used to train the model.	☐	☐
Accuracy is always the primary metric used to measure a model's performance.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Labelling is the process of tagging training data with known values.	☒	☐
You should evaluate a model by using the same data used to train the model.	☐	☒
Accuracy is always the primary metric used to measure a model's performance.	☐	☒

Box 1: Yes

In machine learning, if you have labeled data, that means your data is marked up, or annotated, to show the target, which is the answer you want your machine learning model to predict.

In general, data labeling can refer to tasks that include data tagging, annotation, classification, moderation, transcription, or processing.

Box 2: No

Box 3: No

Accuracy is simply the proportion of correctly classified instances. It is usually the first metric you look at when evaluating a classifier. However, when the test data is unbalanced (where most of the instances belong to one of the classes), or you are more interested in the performance on either one of the classes, accuracy doesn't really capture the effectiveness of a classifier.

Reference:

<https://www.cloudfactory.com/data-labeling-guide> <https://docs.microsoft.com/en-us/azure/machine-learning/studio/evaluate-model-performance>

QUESTION: 35

Which service should you use to extract text, key/value pairs, and table data automatically from scanned documents?

- A. Form Recognizer
- B. Text Analytics
- C. Language Understanding
- D. Custom Vision

Answer(s): A

Explanation:

Accelerate your business processes by automating information extraction. Form Recognizer applies advanced machine learning to accurately extract text, key/ value pairs, and tables from documents. With just a few samples, Form Recognizer tailors its understanding to your documents, both on-premises and in the cloud. Turn forms into usable data at a fraction of the time and cost, so you can focus more time acting on the information rather than compiling it.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/form-recognizer/>

QUESTION: 36

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

The ability to extract subtotals and totals from a receipt is a capability of the service.

Custom Vision
Form Recognizer
Ink Recognizer
Text Analytics

Answer(s): A

Explanation:

Answer Area

The ability to extract subtotals and totals from a receipt is a capability of the service.

Custom Vision
Form Recognizer
Ink Recognizer
Text Analytics

Accelerate your business processes by automating information extraction. Form Recognizer applies advanced machine learning to accurately extract text, key/ value pairs, and tables from documents. With just a few samples, Form Recognizer tailors its understanding to your documents, both on-premises and in the cloud. Turn forms into usable data at a fraction of the time and cost, so you can focus more time acting on the information rather than compiling it.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/form-recognizer/>

QUESTION: 37

You use Azure Machine Learning designer to publish an inference pipeline.

Which two parameters should you use to access the web service? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. the model name
- B. the training endpoint
- C. the authentication key
- D. the REST endpoint

Answer(s): C,D

Explanation:

You can consume a published pipeline in the Published pipelines page. Select a published pipeline and find the REST endpoint of it.

To consume the pipeline, you need:

- The REST endpoint for your service
- The Primary Key for your service

Reference:

<https://docs.microsoft.com/en-in/learn/modules/create-regression-model-azure-machine-learning-designer/deploy-service>

QUESTION: 38

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

From Azure Machine Learning designer, to deploy a real-time inference pipeline as a service for others to consume, you must deploy the model to

- a local web service.
- Azure Container Instances.
- Azure Kubernetes Service (AKS).
- Azure Machine Learning compute.

Answer(s): A

Explanation:

Answer Area

From Azure Machine Learning designer, to deploy a real-time inference pipeline as a service for others to consume, you must deploy the model to

- a local web service.
- Azure Container Instances.
- Azure Kubernetes Service (AKS).
- Azure Machine Learning compute.

To perform real-time inferencing, you must deploy a pipeline as a real-time endpoint.

Real-time endpoints must be deployed to an Azure Kubernetes Service cluster.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-designer#deploy>

QUESTION: 39

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Predicting how many hours of overtime a delivery person will work based on the number of order received is an example of

	▼
classification.	
clustering.	
regression.	

Answer(s): A

Explanation:

Answer Area

Predicting how many hours of overtime a delivery person will work based on the number of order received is an example of

	▼
classification.	
clustering.	
regression.	

In the most basic sense, regression refers to prediction of a numeric target.

Linear regression attempts to establish a linear relationship between one or more independent variables and a numeric outcome, or dependent variable.

You use this module to define a linear regression method, and then train a model using a labeled dataset.

The trained model can then be used to make predictions.

Incorrect Answers:

-Classification is a machine learning method that uses data to determine the category, type, or class of an item or row of data.

-Clustering, in machine learning, is a method of grouping data points into similar clusters. It is also called segmentation.

Over the years, many clustering algorithms have been developed. Almost all clustering algorithms use the features of individual items to find similar items. For example, you might apply clustering to find similar people by demographics. You might use clustering with text analysis to group sentences with similar topics or sentiment.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/algorithm-module-reference/linear-regression>

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/machine-learning-initialize-model-clustering>

QUESTION: 40

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Azure Machine Learning designer provides a drag-and-drop visual canvas to build, test, and deploy machine learning models.	☐	☐
Azure Machine Learning designer enables you to save your progress as a pipeline draft.	☐	☐
Azure Machine Learning designer enables you to include custom JavaScript functions.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Azure Machine Learning designer provides a drag-and-drop visual canvas to build, test, and deploy machine learning models.	☒	☐
Azure Machine Learning designer enables you to save your progress as a pipeline draft.	☒	☐
Azure Machine Learning designer enables you to include custom JavaScript functions.	☐	☒

Box 1: Yes

Azure Machine Learning designer lets you visually connect datasets and modules on an interactive canvas to create machine learning models.

Box 2: Yes

With the designer you can connect the modules to create a pipeline draft.

As you edit a pipeline in the designer, your progress is saved as a pipeline draft.

Box 3: No

Reference:

QUESTION: 41

HOTSPOT (Drag and Drop is not supported)

You have the following dataset.

Household Income	Postal Code	House Price Category
20,000	55555	Low
23,000	20541	Middle
80,000	87960	High

You plan to use the dataset to train a model that will predict the house price categories of houses. What are Household Income and House Price Category? To answer, select the appropriate option in the answer area.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Household Income:

A feature

A label

House Price Category:

A feature

A label

Answer(s): A

Explanation:

Answer Area

Household Income:

A feature

A label

House Price Category:

A feature

A label

Reference:

QUESTION: 42

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Azure Machine Learning designer lets you create machine learning models by

	▼
adding and connecting modules on a visual canvas.	
automatically performing common data preparation tasks.	
automatically selecting an algorithm to build the most accurate model.	
using a code-first notebook experience.	

Answer(s): A

Explanation:

Answer Area

Azure Machine Learning designer lets you create machine learning models by

	▼
adding and connecting modules on a visual canvas.	
automatically performing common data preparation tasks.	
automatically selecting an algorithm to build the most accurate model.	
using a code-first notebook experience.	

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-designer>

QUESTION: 43

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Automated machine learning provides you with the ability to include custom Python scripts in a training pipeline.	☐	☐
Automated machine learning implements machine learning solutions without the need for programming experience.	☐	☐
Automated machine learning provides you with the ability to visually connect datasets and modules on an interactive canvas.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Automated machine learning provides you with the ability to include custom Python scripts in a training pipeline.	☒	☐
Automated machine learning implements machine learning solutions without the need for programming experience.	☒	☐
Automated machine learning provides you with the ability to visually connect datasets and modules on an interactive canvas.	☒	☐

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/how-to-designer-python>

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-automated-ml>

QUESTION: 44

A medical research project uses a large anonymized dataset of brain scan images that are categorized into predefined brain haemorrhage types.

You need to use machine learning to support early detection of the different brain haemorrhage types in the images before the images are reviewed by a person.

This is an example of which type of machine learning?

- A. clustering
- B. regression
- C. classification

Answer(s): C

Reference:

<https://docs.microsoft.com/en-us/learn/modules/create-classification-model-azure-machine-learning-designer/introduction>

QUESTION: 45

When training a model, why should you randomly split the rows into separate subsets?

- A. to train the model twice to attain better accuracy
- B. to train multiple models simultaneously to attain better performance
- C. to test the model by using data that was not used to train the model

Answer(s): C

QUESTION: 46

You are evaluating whether to use a basic workspace or an enterprise workspace in Azure Machine Learning.

What are two tasks that require an enterprise workspace? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Use a graphical user interface (GUI) to run automated machine learning experiments.
- B. Create a compute instance to use as a workstation.
- C. Use a graphical user interface (GUI) to define and run machine learning experiments from Azure Machine Learning designer.
- D. Create a dataset from a comma-separated value (CSV) file.

Answer(s): A,C

Explanation:

Note: Enterprise workspaces are no longer available as of September 2020. The basic workspace now has all the functionality of the enterprise workspace.

Reference:

<https://www.azure.cn/en-us/pricing/details/machine-learning/>

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-workspace>

QUESTION: 47

You need to predict the income range of a given customer by using the following dataset.

First Name	Last Name	Age	Education Level	Income Range
Orlando	Gee	45	University	25,000-50,000
Keith	Harris	36	High school	25,000-50,000
Donna	Carreras	52	University	50,000-75,000
Janet	Gates	21	University	75,000-100,000
Lucy	Harrington	68	High school	50,000-75,000

Which two fields should you use as features? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Education Level
- B. Last Name
- C. Age
- D. Income Range
- E. First Name

Answer(s): A,C

Explanation:

First Name, Last Name, Age and Education Level are features. Income range is a label (what you want to predict). First Name and Last Name are irrelevant in that they have no bearing on income. Age and Education level are the features you should use.

QUESTION: 48

You are building a tool that will process images from retail stores and identify the products of competitors. The solution will use a custom model.

Which Azure Cognitive Services service should you use?

- A. Custom Vision
- B. Form Recognizer
- C. Face
- D. Computer Vision

Answer(s): A

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/overview>

QUESTION: 49

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Organizing documents into groups based on similarities of the text contained in the documents is an example of clustering.	☐	☐
Grouping similar patients based on symptoms and diagnostic test results is an example of clustering.	☐	☐
Predicting whether a person will develop mild, moderate, or severe allergy symptoms based on pollen count is an example of clustering.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Organizing documents into groups based on similarities of the text contained in the documents is an example of clustering.	☒	☐
Grouping similar patients based on symptoms and diagnostic test results is an example of clustering.	☒	☐
Predicting whether a person will develop mild, moderate, or severe allergy symptoms based on pollen count is an example of clustering.	☐	☒

Clustering is a machine learning task that is used to group instances of data into clusters that contain similar characteristics. Clustering can also be used to identify relationships in a dataset

Regression is a machine learning task that is used to predict the value of the label from a set of related features.

Reference:

<https://docs.microsoft.com/en-us/dotnet/machine-learning/resources/tasks>

QUESTION: 50

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
A validation set includes the set of input examples that will be used to train a mode.	☐	☐
A validation set can be used to determine how well a model predicts labels.	☐	☐
A validation set can be used to verify that all the training data was used to train the model.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
A validation set includes the set of input examples that will be used to train a mode.	☐	☒
A validation set can be used to determine how well a model predicts labels.	☒	☐
A validation set can be used to verify that all the training data was used to train the model.	☐	☒

Box 1: No

The validation dataset is different from the test dataset that is held back from the training of the model.

Box 2: Yes

A validation dataset is a sample of data that is used to give an estimate of model skill while tuning model's hyperparameters.

Box 3: No

The Test Dataset, not the validation set, used for this. The Test Dataset is a sample of data used to provide an unbiased evaluation of a final model fit on the training dataset.

Reference:

<https://machinelearningmastery.com/difference-test-validation-datasets/>

QUESTION: 51

What are two metrics that you can use to evaluate a regression model? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. coefficient of determination (R2)
- B. F1 score
- C. root mean squared error (RMSE)
- D. area under curve (AUC)
- E. balanced accuracy

Answer(s): A,C

Explanation:

A: R-squared (R2), or Coefficient of determination represents the predictive power of the model as a value between -inf and 1.00. 1.00 means there is a perfect fit, and the fit can be arbitrarily poor so the scores can be negative.

C: RMS-loss or Root Mean Squared Error (RMSE) (also called Root Mean Square Deviation, RMSD), measures the difference between values predicted by a model and the values observed from the environment that is being modeled.

Incorrect Answers:

B: F1 score also known as balanced F-score or F-measure is used to evaluate a classification model.

D: aucROC or area under the curve (AUC) is used to evaluate a classification model.

Reference:

<https://docs.microsoft.com/en-us/dotnet/machine-learning/resources/metrics>

QUESTION: 52

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Predicting how many vehicles will travel across a bridge on a given day is an example of

	▼
classification.	
clustering.	
regression.	

Answer(s): A

Explanation:

Answer Area

Predicting how many vehicles will travel across a bridge on a given day is an example of

classification.
clustering.
regression.

Regression is a machine learning task that is used to predict the value of the label from a set of related features.

Reference:

<https://docs.microsoft.com/en-us/dotnet/machine-learning/resources/tasks>

QUESTION: 53

DRAG DROP (Drag and Drop is not supported)

You need to use Azure Machine Learning designer to build a model that will predict automobile prices.

Which type of modules should you use to complete the model? To answer, drag the appropriate modules to the correct locations. Each module may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

Note: Each correct selection is worth one point.

Select and Place:

Modules

Convert to CSV

K-Means Clustering

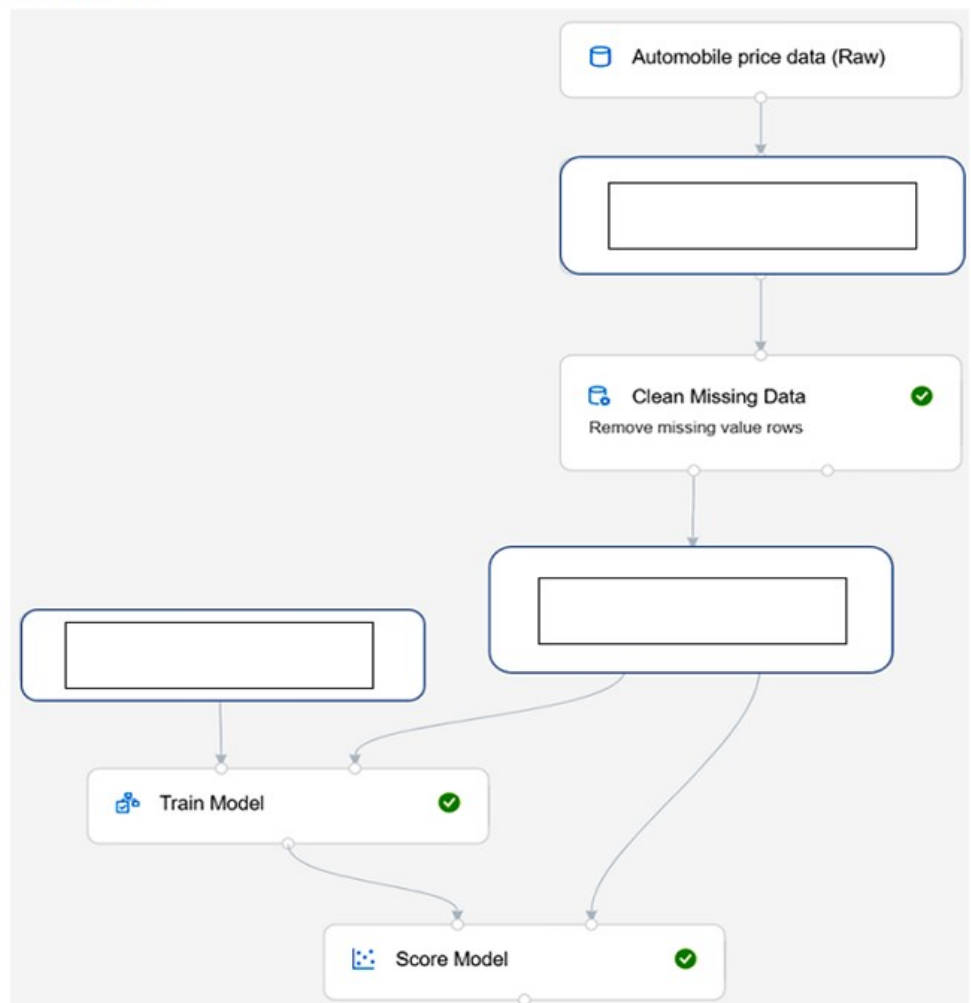
Linear Regression

Split Data

Select Columns in Dataset

Summarize Data

Answer Area



Answer(s): A

Explanation:

Modules

Convert to CSV

K-Means Clustering

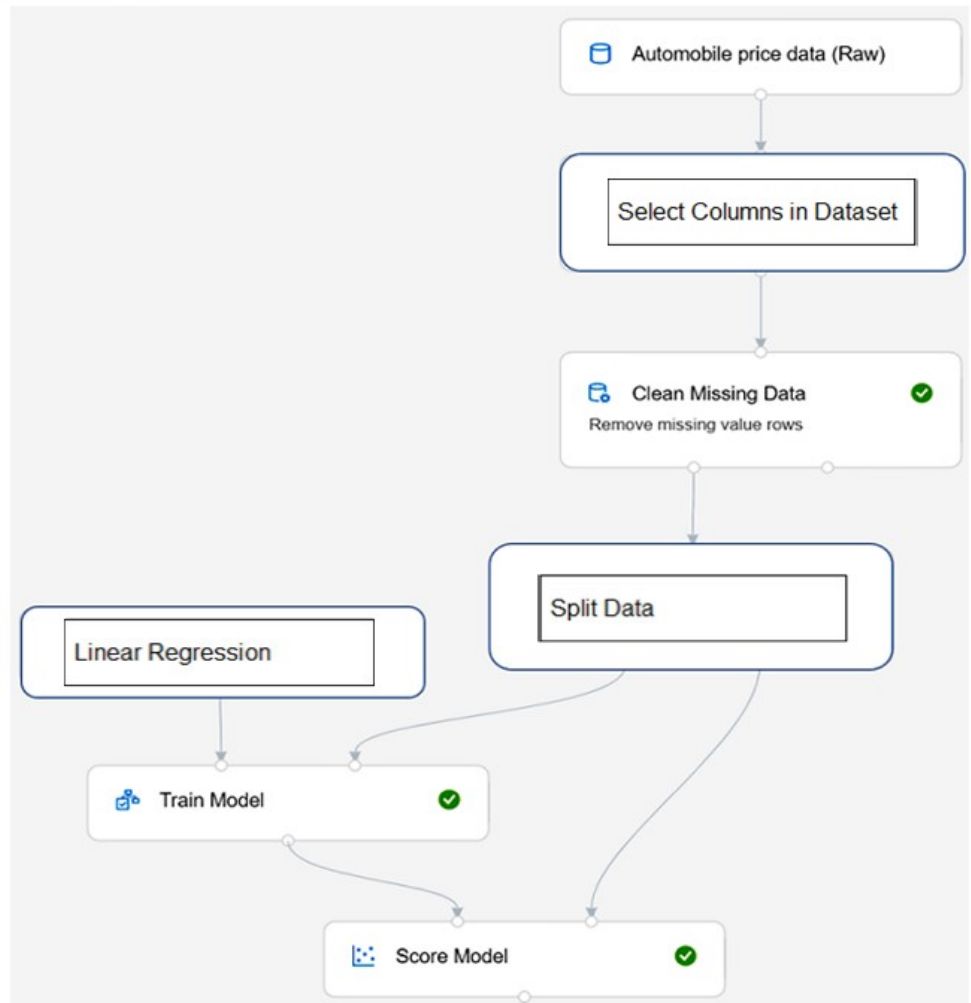
Linear Regression

Split Data

Select Columns in Dataset

Summarize Data

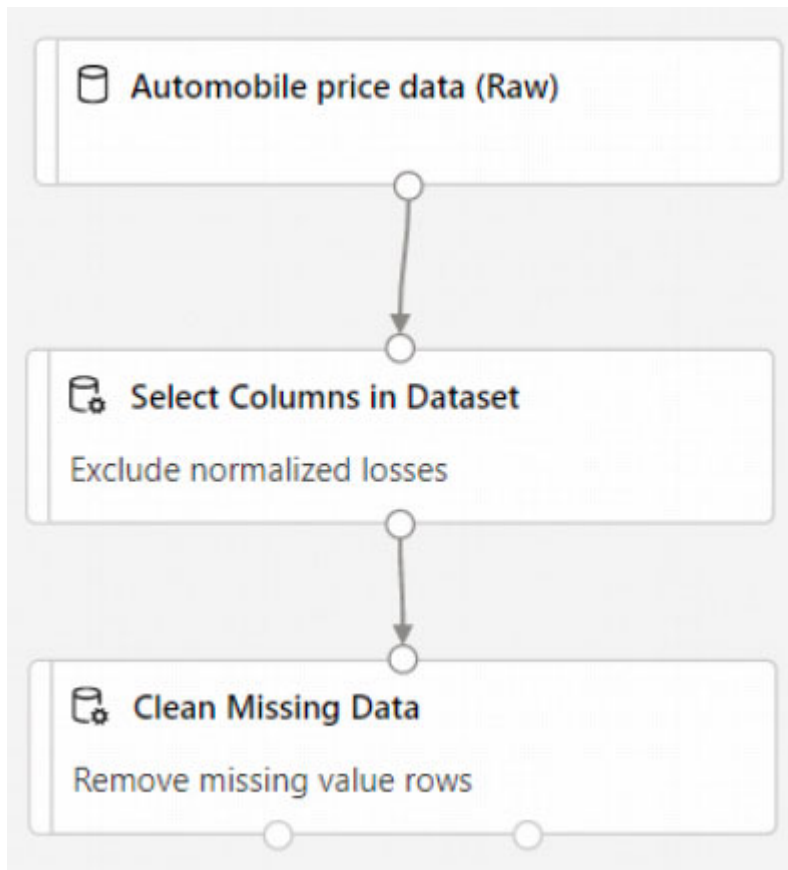
Answer Area



Box 1: Select Columns in Dataset

For Columns to be cleaned, choose the columns that contain the missing values you want to change. You can choose multiple columns, but you must use the same replacement method in all selected columns.

Example:



Box 2: Split data

Splitting data is a common task in machine learning. You will split your data into two separate datasets. One dataset will train the model and the other will test how well the model performed.

Box 3: Linear regression

Because you want to predict price, which is a number, you can use a regression algorithm. For this example, you use a linear regression model.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/tutorial-designer-automobile-price-train-score>

QUESTION: 54

Which type of machine learning should you use to identify groups of people who have similar purchasing habits?

- A. classification
- B. regression
- C. clustering

Answer(s): C

Explanation:

Clustering is a machine learning task that is used to group instances of data into clusters that contain similar characteristics. Clustering can also be used to identify relationships in a dataset

Reference:

<https://docs.microsoft.com/en-us/dotnet/machine-learning/resources/tasks>

QUESTION: 55

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

	▼
Classification	
Clustering	
Regression	

models can be used to predict the sale price of auctioned items.

Answer(s): A

Explanation:

Answer Area

	▼
Classification	
Clustering	
Regression	

models can be used to predict the sale price of auctioned items.

Regression is a machine learning task that is used to predict the value of the label from a set of related features.

Reference:

<https://docs.microsoft.com/en-us/dotnet/machine-learning/resources/tasks>

QUESTION: 56

Which metric can you use to evaluate a classification model?

- A. true positive rate
- B. mean absolute error (MAE)
- C. coefficient of determination (R²)
- D. root mean squared error (RMSE)

Answer(s): A

Explanation:

What does a good model look like?

An ROC curve that approaches the top left corner with 100% true positive rate and 0% false positive rate will be the best model. A random model would display as a flat line from the bottom left to the top right corner. Worse than random would dip below the $y=x$ line.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/how-to-understand-automated-ml#classification>

QUESTION: 57

Which two components can you drag onto a canvas in Azure Machine Learning designer? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. dataset
- B. compute
- C. pipeline
- D. module

Answer(s): A,D

Explanation:

You can drag-and-drop datasets and modules onto the canvas.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-designer>

QUESTION: 58

You need to create a training dataset and validation dataset from an existing dataset.

Which module in the Azure Machine Learning designer should you use?

- A. Select Columns in Dataset
- B. Add Rows
- C. Split Data
- D. Join Data

Answer(s): C

Explanation:

A common way of evaluating a model is to divide the data into a training and test set by using Split Data, and then validate the model on the training data.

Use the Split Data module to divide a dataset into two distinct sets.

The studio currently supports training/validation data splits

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/how-to-configure-cross-validation-data-splits>

QUESTION: 59

DRAG DROP (Drag and Drop is not supported)

Match the types of machine learning to the appropriate scenarios.

To answer, drag the appropriate machine learning type from the column on the left to its scenario on the right. Each machine learning type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Learning Types	Answer Area
Classification	Learning Type Predict how many minutes late a flight will arrive based on the amount of snowfall at an airport.
Clustering	Learning Type Segment customers into different groups to support a marketing department.
Regression	Learning Type Predict whether a student will complete a university course.

Answer(s): A

Explanation:

Learning Types	Answer Area
Classification	Regression Predict how many minutes late a flight will arrive based on the amount of snowfall at an airport.
Clustering	Clustering Segment customers into different groups to support a marketing department.
Regression	Classification Predict whether a student will complete a university course.

Box 1: Regression

In the most basic sense, regression refers to prediction of a numeric target.

Linear regression attempts to establish a linear relationship between one or more independent variables and a numeric outcome, or dependent variable.

You use this module to define a linear regression method, and then train a model using a labeled dataset. The trained model can then be used to make predictions.

Box 2: Clustering

Clustering, in machine learning, is a method of grouping data points into similar clusters. It is also called segmentation.

Over the years, many clustering algorithms have been developed. Almost all clustering algorithms use the features of individual items to find similar items. For example, you might apply clustering to find similar

people by demographics. You might use clustering with text analysis to group sentences with similar topics or sentiment.

Box 3: Classification

Two-class classification provides the answer to simple two-choice questions such as Yes/No or True/False.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/linear-regression>

QUESTION: 60

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

	▼
Accuracy	
Confidence	
Root Mean Square Error	
Sentiment	

is the calculated probability of a correct image classification.

Answer(s): A

Explanation:

Answer Area

	▼
Accuracy	
Confidence	
Root Mean Square Error	
Sentiment	

is the calculated probability of a correct image classification.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/getting-started-build-a-classifier>

QUESTION: 61

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Ensuring an AI system does not provide a prediction when important fields contain unusual or missing values is

	▼
an inclusiveness	
a privacy and security	
a reliability and safety	
a transparency	

 principle for responsible AI.

Answer(s): A

Explanation:

Answer Area

Ensuring an AI system does not provide a prediction when important fields contain unusual or missing values is

	▼
an inclusiveness	
a privacy and security	
a reliability and safety	
a transparency	

 principle for responsible AI.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>

QUESTION: 62

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Ensuring that the numeric variables in training data are on a similar scale is an example of

	▼
data ingestion.	
feature engineering.	
feature selection.	
model training.	

Answer(s): A

Explanation:

Answer Area

Ensuring that the numeric variables in training data are on a similar scale is an example of

	▼
data ingestion.	
feature engineering.	
feature selection.	
model training.	

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-science-process/create-features>

QUESTION: 63

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Assigning classes to images before training a classification model is an example of

	▼
evaluation.	
feature engineering	
hyperparameter tuning.	
labeling.	

Answer(s): A

Explanation:

Answer Area

Assigning classes to images before training a classification model is an example of

	▼
evaluation.	
feature engineering	
hyperparameter tuning.	
labeling.	

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/how-to-label-data>

QUESTION: 64

HOTSPOT (Drag and Drop is not supported)

You have an Azure Machine Learning model that predicts product quality. The model has a training dataset that contains 50,000 records. A sample of the data is shown in the following table.

Date	Time	Mass (kg)	Temperature (C)	Quality Test
26/02/2021	15:31:07	2.108	62.5	Pass
26/02/2021	15:31:39	2.099	62.4	Pass
26/02/2021	02:32:21	2.098	66.4	Fail

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Mass (kg) is a feature.	☐	☐
Quality Test is a label.	☐	☐
Temperature (C) is a label.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Mass (kg) is a feature.	☒	☐
Quality Test is a label.	☒	☐
Temperature (C) is a label.	☐	☒

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/component-reference/filter-based-feature-selection>

QUESTION: 65

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
You train a regression model by using unlabeled data.	☐	☐
The classification technique is used to predict sequential numerical data over time.	☐	☐
Grouping items by their common characteristics is an example of clustering.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
You train a regression model by using unlabeled data.	☐	☒
The classification technique is used to predict sequential numerical data over time.	☐	☒
Grouping items by their common characteristics is an example of clustering.	☒	☐

Reference:

<https://docs.microsoft.com/en-us/learn/modules/create-regression-model-azure-machine-learning-designer/5-create-training-pipeline> <https://docs.microsoft.com/en-us/learn/modules/create-classification-model-azure-machine-learning-designer/introduction> <https://docs.microsoft.com/en-us/learn/modules/create-clustering-model-azure-machine-learning-designer/1-introduction>

QUESTION: 66

Which two actions are performed during the data ingestion and data preparation stage of an Azure Machine Learning process? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. Calculate the accuracy of the model.
- B. Score test data by using the model.
- C. Combine multiple datasets.
- D. Use the model for real-time predictions.
- E. Remove records that have missing values.

Answer(s): C,E

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/concept-data-ingestion>

<https://docs.microsoft.com/en-us/azure/architecture/data-science-process/prepare-data>

QUESTION: 67

You need to predict the animal population of an area.
Which Azure Machine Learning type should you use?

- A. regression
- B. clustering
- C. classification

Answer(s): A

Explanation:

Regression is a supervised machine learning technique used to predict numeric values.

Reference:

<https://docs.microsoft.com/en-us/learn/modules/create-regression-model-azure-machine-learning-designer/1-introduction>

QUESTION: 68

Which two languages can you use to write custom code for Azure Machine Learning designer? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Python
- B. R
- C. C#
- D. Scala

Answer(s): A,B

Explanation:

Use Azure Machine Learning designer for customizing using Python and R code.

Reference:

<https://azure.microsoft.com/en-us/services/machine-learning/designer/#features>

QUESTION: 69

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
For a regression model, labels must be numeric.	☐	☐
For a clustering model, labels must be used.	☐	☐
For a classification model, labels must be numeric.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
For a regression model, labels must be numeric.	☒	☐
For a clustering model, labels must be used.	☐	☒
For a classification model, labels must be numeric.	☐	☒

Box 1: Yes

For regression problems, the label column must contain numeric data that represents the response variable. Ideally the numeric data represents a continuous scale.

Box 2: No

K-Means Clustering

Because the K-means algorithm is an unsupervised learning method, a label column is optional.

If your data includes a label, you can use the label values to guide selection of the clusters and optimize the model.

If your data has no label, the algorithm creates clusters representing possible categories, based solely on the data.

Box 3: No

For classification problems, the label column must contain either categorical values or discrete values. Some examples might be a yes/no rating, a disease classification code or name, or an income group. If you pick a noncategorical column, the component will return an error during training.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/component-reference/train-model>

<https://docs.microsoft.com/en-us/azure/machine-learning/component-reference/k-means-clustering>

QUESTION: 70

Your company wants to build a recycling machine for bottles. The recycling machine must automatically identify bottles of the correct shape and reject all other items.

Which type of AI workload should the company use?

- A. anomaly detection
- B. conversational AI
- C. computer vision
- D. natural language processing

Answer(s): C

Explanation:

Azure's Computer Vision service gives you access to advanced algorithms that process images and return information based on the visual features you're interested in. For example, Computer Vision can determine whether an image contains adult content, find specific brands or objects, or find human faces.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview>

QUESTION: 71

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
When creating an object detection model in the Custom Vision service, you must choose a classification type of either Multilabel or Multiclass .	☐	☐
You can create an object detection model in the Custom Vision service to find the location of content within an image.	☐	☐
When creating an object detection model in the Custom Vision service, you can select from a set of predefined domains.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
When creating an object detection model in the Custom Vision service, you must choose a classification type of either Multilabel or Multiclass .	☐	☒
You can create an object detection model in the Custom Vision service to find the location of content within an image.	☒	☐
When creating an object detection model in the Custom Vision service, you can select from a set of predefined domains.	☒	☐

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/get-started-build-detector>

QUESTION: 72

In which two scenarios can you use the Form Recognizer service? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Extract the invoice number from an invoice.
- B. Translate a form from French to English.
- C. Find image of product in a catalog.
- D. Identify the retailer from a receipt.

Answer(s): A,D

Reference:

<https://azure.microsoft.com/en-gb/services/cognitive-services/form-recognizer/#features>

QUESTION: 73

HOTSPOT (Drag and Drop is not supported)

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

Counting the number of animals in an area based on a video feed is an example of

	▼
forecasting.	
computer vision.	
conversational AI.	
anomaly detection.	

Answer(s): A

Explanation:

Answer Area

Counting the number of animals in an area based on a video feed is an example of

	▼
forecasting.	
computer vision.	
conversational AI.	
anomaly detection.	

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview>

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/intro-to-spatial-analysis-public-preview>

QUESTION: 74

HOTSPOT (Drag and Drop is not supported)

You have a database that contains a list of employees and their photos.

You are tagging new photos of the employees.

For each of the following statements select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
The Face service can be used to perform facial recognition for employees	☐	☐
The Face service will be more accurate if you provide more sample photos of each employee from different angles.	☐	☐
If an employee is wearing sunglasses, the Face service will always fail to recognize the employee.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
The Face service can be used to perform facial recognition for employees	☒	☐
The Face service will be more accurate if you provide more sample photos of each employee from different angles.	☒	☐
If an employee is wearing sunglasses, the Face service will always fail to recognize the employee.	☐	☒

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/face/overview> <https://docs.microsoft.com/en-us/azure/cognitive-services/face/concepts/face-detection>

QUESTION: 75

You need to develop a mobile app for employees to scan and store their expenses while travelling. Which type of computer vision should you use?

- A. semantic segmentation
- B. image classification
- C. object detection
- D. optical character recognition (OCR)

Answer(s): D

Explanation:

Azure's Computer Vision API includes Optical Character Recognition (OCR) capabilities that extract printed or handwritten text from images. You can extract text from images, such as photos of license plates or containers with serial numbers, as well as from documents - invoices, bills, financial reports, articles, and more.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-recognizing-text>

QUESTION: 76

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
The Custom Vision service can be used to detect objects in an image.	☐	☐
The Custom Vision service requires that you provide your own data to train the model.	☐	☐
The Custom Vision service can be used to analyze video files.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
The Custom Vision service can be used to detect objects in an image.	☒	☐
The Custom Vision service requires that you provide your own data to train the model.	☒	☐
The Custom Vision service can be used to analyze video files.	☐	☒

Box 1: Yes

Custom Vision functionality can be divided into two features. Image classification applies one or more labels to an image. Object detection is similar, but it also returns the coordinates in the image where the applied label(s) can be found.

Box 2: Yes

The Custom Vision service uses a machine learning algorithm to analyze images. You, the developer, submit groups of images that feature and lack the characteristics in question. You label the images yourself at the time of submission. Then, the algorithm trains to this data and calculates its own accuracy by testing itself on those same images.

Box 3: No

Custom Vision service can be used only on graphic files.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/Custom-Vision-Service/overview>

QUESTION: 77

You are processing photos of runners in a race.

You need to read the numbers on the runners' shirts to identity the runners in the photos.

Which type of computer vision should you use?

- A. facial recognition
- B. optical character recognition (OCR)
- C. image classification
- D. object detection

Answer(s): B

Explanation:

Optical character recognition (OCR) allows you to extract printed or handwritten text from images and documents.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview-ocr>

QUESTION: 78

DRAG DROP (Drag and Drop is not supported)

Match the types of machine learning to the appropriate scenarios.

To answer, drag the appropriate machine learning type from the column on the left to its scenario on the right. Each machine learning type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Machine Learning Types

- Facial detection
- Facial recognition
- Image classification
- Object detection
- Optical character recognition (OCR)
- Semantic segmentation

Answer Area

- Machine Learning Type Separate images of polar bears and brown bears.
- Machine Learning Type Determine the location of a bear in a photo.
- Machine Learning Type Determine which pixels in an image are part of a bear.

Answer(s): A

Explanation:

Machine Learning Types

- Facial detection
- Facial recognition
- Image classification
- Object detection
- Optical character recognition (OCR)
- Semantic segmentation

Answer Area

- Image classification Separate images of polar bears and brown bears.
- Object detection Determine the location of a bear in a photo.
- Semantic segmentation Determine which pixels in an image are part of a bear.

Box 1: Image classification

Image classification is a supervised learning problem: define a set of target classes (objects to identify in images), and train a model to recognize them using labeled example photos.

Box 2: Object detection

Object detection is a computer vision problem.

While closely related to image classification, object detection performs image classification at a more granular scale. Object detection both locates and categorizes entities within images.

Box 3: Semantic Segmentation

Semantic segmentation achieves fine-grained inference by making dense predictions inferring labels for every pixel, so that each pixel is labeled with the class of its enclosing object or region.

Reference:

<https://developers.google.com/machine-learning/practica/image-classification>

<https://docs.microsoft.com/en-us/dotnet/machine-learning/tutorials/object-detection-model-builder>

<https://nanonets.com/blog/how-to-do-semantic-segmentation-using-deep-learning/>

QUESTION: 79

You use drones to identify where weeds grow between rows of crops to send an instruction for the removal of the weeds.

This is an example of which type of computer vision?

- A. object detection
- B. optical character recognition (OCR)
- C. scene segmentation

Answer(s): A

Explanation:

Object detection is similar to tagging, but the API returns the bounding box coordinates for each tag applied. For example, if an image contains a dog, cat and person, the Detect operation will list those objects together with their coordinates in the image.

Incorrect Answers:

B: Optical character recognition (OCR) allows you to extract printed or handwritten text from images and documents.

C: Scene segmentation determines when a scene changes in video based on visual cues. A scene depicts a single event and it's composed by a series of consecutive shots, which are semantically related.

Reference:

<https://docs.microsoft.com/en-us/ai-builder/object-detection-overview> <https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview-ocr> <https://docs.microsoft.com/en-us/azure/azure-video-analyzer/video-analyzer-for-media-docs/video-indexer-overview>

QUESTION: 80

DRAG DROP (Drag and Drop is not supported)

Match the facial recognition tasks to the appropriate questions.

To answer, drag the appropriate task from the column on the left to its question on the right. Each task may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Tasks

grouping
identification
similarity
verification

Answer Area

Task	Do two images of a face belong to the same person?
Task	Does this person look like other people?
Task	Do all the faces belong together?
Task	Who is this person in this group of people?

Answer(s): A

Explanation:

Tasks

grouping
identification
similarity
verification

Answer Area

verification	Do two images of a face belong to the same person?
similarity	Does this person look like other people?
grouping	Do all the faces belong together?
identification	Who is this person in this group of people?

Box 1: verification

Face verification: Check the likelihood that two faces belong to the same person and receive a confidence score.

Box 2: similarity

Box 3: Grouping

Box 4: identification

Face detection: Detect one or more human faces along with attributes such as: age, emotion, pose, smile, and facial hair, including 27 landmarks for each face in the image.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/face/#features>

QUESTION: 81

DRAG DROP (Drag and Drop is not supported)

Match the types of computer vision workloads to the appropriate scenarios.

To answer, drag the appropriate workload type from the column on the left to its scenario on the right. Each workload type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Workloads Types

Facial recognition
Image classification
Object detection
Optical character recognition (OCR)

Answer Area

Workload Type	Identify celebrities in images.
Workload Type	Extract movie title names from movie poster images.
Workload Type	Locate vehicles in images.

Answer(s): A

Explanation:

Workloads Types

Facial recognition
Image classification
Object detection
Optical character recognition (OCR)

Answer Area

Facial recognition	Identify celebrities in images.
Optical character recognition (OCR)	Extract movie title names from movie poster images.
Object detection	Locate vehicles in images.

Box 1: Facial recognition

Face detection that perceives faces and attributes in an image; person identification that matches an individual in your private repository of up to 1 million people; perceived emotion recognition that detects a range of facial expressions like happiness, contempt, neutrality, and fear; and recognition and grouping of similar faces in images.

Box 2: OCR

Box 3: Objection detection

Object detection is similar to tagging, but the API returns the bounding box coordinates (in pixels) for each object found. For example, if an image contains a dog, cat and person, the Detect operation will list those objects together with their coordinates in the image. You can use this functionality to process the relationships between the objects in an image. It also lets you determine whether there are multiple instances of the same tag in an image.

The Detect API applies tags based on the objects or living things identified in the image. There is currently no formal relationship between the tagging taxonomy and the object detection taxonomy. At a conceptual level, the Detect API only finds objects and living things, while the Tag API can also include contextual terms like "indoor", which can't be localized with bounding boxes.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/face/> <https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-object-detection>

QUESTION: 82

You need to determine the location of cars in an image so that you can estimate the distance between the cars.

Which type of computer vision should you use?

- A. optical character recognition (OCR)
- B. object detection
- C. image classification
- D. face detection

Answer(s): B

Explanation:

Object detection is similar to tagging, but the API returns the bounding box coordinates (in pixels) for each object found. For example, if an image contains a dog, cat and person, the Detect operation will list those objects together with their coordinates in the image. You can use this functionality to process the relationships between the objects in an image. It also lets you determine whether there are multiple instances of the same tag in an image.

The Detect API applies tags based on the objects or living things identified in the image. There is currently no formal relationship between the tagging taxonomy and the object detection taxonomy. At a conceptual level, the Detect API only finds objects and living things, while the Tag API can also include contextual terms like "indoor", which can't be localized with bounding boxes.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-object-detection>

QUESTION: 83

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

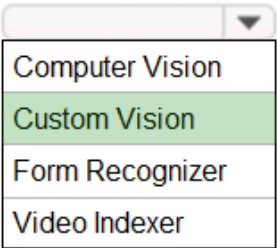
You can use the service to train an object detection model by using your own images.

Computer Vision
Custom Vision
Form Recognizer
Video Indexer

Answer(s): A

Explanation:

Answer Area

You can use the  service to train an object detection model by using your own images.

Computer Vision
Custom Vision
Form Recognizer
Video Indexer

Azure Custom Vision is a cognitive service that lets you build, deploy, and improve your own image classifiers. An image classifier is an AI service that applies labels (which represent classes) to images, according to their visual characteristics. Unlike the Computer Vision service, Custom Vision allows you to specify the labels to apply.

Note: The Custom Vision service uses a machine learning algorithm to apply labels to images. You, the developer, must submit groups of images that feature and lack the characteristics in question. You label the images yourself at the time of submission. Then the algorithm trains to this data and calculates its own accuracy by testing itself on those same images. Once the algorithm is trained, you can test, retrain, and eventually use it to classify new images according to the needs of your app. You can also export the model itself for offline use.

Incorrect Answers:

Computer Vision:

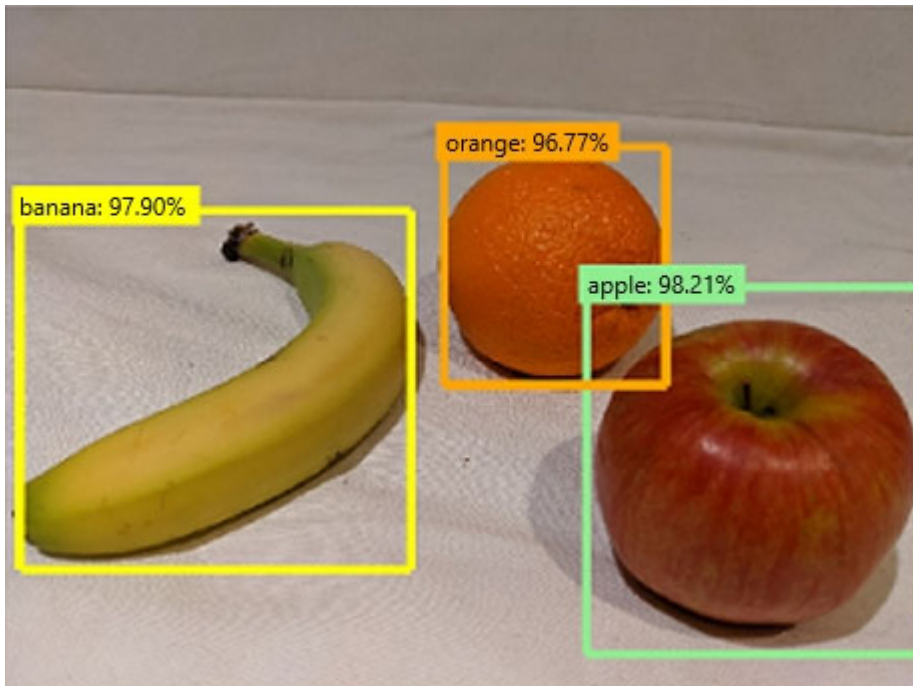
Azure's Computer Vision service provides developers with access to advanced algorithms that process images and return information based on the visual features you're interested in. For example, Computer Vision can determine whether an image contains adult content, find specific brands or objects, or find human faces.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/home>

QUESTION: 84

You send an image to a Computer Vision API and receive back the annotated image shown in the exhibit.



Which type of computer vision was used?

- A. object detection
- B. face detection
- C. optical character recognition (OCR)
- D. image classification

Answer(s): A

Explanation:

Object detection is similar to tagging, but the API returns the bounding box coordinates (in pixels) for each object found. For example, if an image contains a dog, cat and person, the Detect operation will list those objects together with their coordinates in the image. You can use this functionality to process the relationships between the objects in an image. It also lets you determine whether there are multiple instances of the same tag in an image.

The Detect API applies tags based on the objects or living things identified in the image. There is currently no formal relationship between the tagging taxonomy and the object detection taxonomy. At a conceptual level, the Detect API only finds objects and living things, while the Tag API can also include contextual terms like "indoor", which can't be localized with bounding boxes.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-object-detection>

QUESTION: 85

What are two tasks that can be performed by using the Computer Vision service? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Train a custom image classification model.
- B. Detect faces in an image.
- C. Recognize handwritten text.
- D. Translate the text in an image between languages.

Answer(s): B,C

Explanation:

B: Azure's Computer Vision service provides developers with access to advanced algorithms that process images and return information based on the visual features you're interested in. For example, Computer Vision can determine whether an image contains adult content, find specific brands or objects, or find human faces.

C: Computer Vision includes Optical Character Recognition (OCR) capabilities. You can use the new Read API to extract printed and handwritten text from images and documents.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/home>

QUESTION: 86

What is a use case for classification?

- A. predicting how many cups of coffee a person will drink based on how many hours the person slept the previous night.
- B. analyzing the contents of images and grouping images that have similar colors
- C. predicting whether someone uses a bicycle to travel to work based on the distance from home to work
- D. predicting how many minutes it will take someone to run a race based on past race times

Answer(s): C

Explanation:

Two-class classification provides the answer to simple two-choice questions such as Yes/No or True/False.

Incorrect Answers:

- A: This is Regression.
- B: This is Clustering.
- D: This is Regression.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/algorithm-module-reference/linear-regression>
<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/machine-learning-initialize-model-clustering>

QUESTION: 87

What are two tasks that can be performed by using computer vision? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Predict stock prices.
- B. Detect brands in an image.
- C. Detect the color scheme in an image
- D. Translate text between languages.
- E. Extract key phrases.

Answer(s): B,C

Explanation:

B: Identify commercial brands in images or videos from a database of thousands of global logos. You can use this feature, for example, to discover which brands are most popular on social media or most prevalent in media product placement.

C: Analyze color usage within an image. Computer Vision can determine whether an image is black & white or color and, for color images, identify the dominant and accent colors.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview>

QUESTION: 88

You need to build an image tagging solution for social media that tags images of your friends automatically. Which Azure Cognitive Services service should you use?

- A. Face
- B. Form Recognizer
- C. Text Analytics
- D. Computer Vision

Answer(s): A

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/face/overview> <https://docs.microsoft.com/en-us/azure/cognitive-services/face/face-api-how-to-topics/howtodetectfacesinimage>

QUESTION: 89

In which two scenarios can you use the Form Recognizer service? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Identify the retailer from a receipt
- B. Translate from French to English

- C. Extract the invoice number from an invoice
- D. Find images of products in a catalog

Answer(s): A,C

Reference:

<https://docs.microsoft.com/en-us/azure/applied-ai-services/form-recognizer/overview?tabs=v2-1>

QUESTION: 90

DRAG DROP (Drag and Drop is not supported)

Match the facial recognition tasks to the appropriate questions.

To answer, drag the appropriate task from the column on the left to its question on the right.

Each task may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Tasks	Answer Area
grouping	Task Do two images of a face belong to the same person?
identification	Task Does this person look like other people?
similarity	Task Who is this person in this group of people?
verification	

Answer(s): A

Explanation:

Tasks	Answer Area
grouping	verification Do two images of a face belong to the same person?
identification	similarity Does this person look like other people?
similarity	identification Who is this person in this group of people?
verification	

Box 1: verification

Identity verification

Modern enterprises and apps can use the Face identification and Face verification operations to verify that a user is who they claim to be.

Box 2: similarity

The Find Similar operation does face matching between a target face and a set of candidate faces, finding a smaller set of faces that look similar to the target face.

This is useful for doing a face search by image.

The service supports two working modes, matchPerson and matchFace. The matchPerson mode returns similar faces after filtering for the same person by using the Verify API. The matchFace mode ignores the same-person filter. It returns a list of similar candidate faces that may or may not belong to the same person.

Box 3: identification

Face identification can address "one-to-many" matching of one face in an image to a set of faces in a secure repository. Match candidates are returned based on how closely their face data matches the query face. This scenario is used in granting building or airport access to a certain group of people or verifying the user of a device.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/face/overview>

QUESTION: 91

Which Computer Vision feature can you use to generate automatic captions for digital photographs?

- A. Recognize text.
- B. Identify the areas of interest.
- C. Detect objects.
- D. Describe the images.

Answer(s): D

Explanation:

Describe images with human-readable language

Computer Vision can analyze an image and generate a human-readable phrase that describes its contents. The algorithm returns several descriptions based on different visual features, and each description is given a confidence score. The final output is a list of descriptions ordered from highest to lowest confidence. The image description feature is part of the Analyze Image API.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-describing-images>

QUESTION: 92

Which service should you use to extract text, key/value pairs, and table data automatically from scanned documents?

- A. Custom Vision
- B. Face

C. Form Recognizer

D. Language

Answer(s): C

Explanation:

Form Recognizer applies advanced machine learning to accurately extract text, key-value pairs, tables, and structures from documents.

Reference:

<https://azure.microsoft.com/en-us/services/form-recognizer/>

QUESTION: 93

HOTSPOT (Drag and Drop is not supported)

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

Object detection
Facial recognition
Image classification
Optical character recognition (OCR)

extracts text from handwritten documents.

Answer(s): A

Explanation:

Answer Area

Object detection
Facial recognition
Image classification
Optical character recognition (OCR)

extracts text from handwritten documents.

Handwriting OCR (optical character recognition) is the process of automatically extracting handwritten information from paper, scans and other low-quality digital documents.

Reference:

<https://vidado.ai/handwriting-ocr>

QUESTION: 94

You are developing a solution that uses the Text Analytics service.
You need to identify the main talking points in a collection of documents.
Which type of natural language processing should you use?

- A. entity recognition
- B. key phrase extraction
- C. sentiment analysis
- D. language detection

Answer(s): B

Explanation:

Broad entity extraction: Identify important concepts in text, including key
Key phrase extraction/ Broad entity extraction: Identify important concepts in text, including key phrases
and named entities such as people, places, and organizations.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 95

In which two scenarios can you use speech recognition? Each correct answer presents a complete solution.
Note: Each correct selection is worth one point.

- A. an in-car system that reads text messages aloud
- B. providing closed captions for recorded or live videos
- C. creating an automated public address system for a train station
- D. creating a transcript of a telephone call or meeting

Answer(s): B,D

Reference:

<https://azure.microsoft.com/en-gb/services/cognitive-services/speech-to-text/#features>

QUESTION: 96

HOTSPOT (Drag and Drop is not supported)
To complete the sentence, select the appropriate option in the answer area.
Hot Area:

Answer Area

While presenting at a conference, your session is transcribed into subtitles for the audience. This is an example of

	▼
sentiment analysis.	
speech recognition.	
speech synthesis.	
translation.	

Answer(s): A

Explanation:

Answer Area

While presenting at a conference, your session is transcribed into subtitles for the audience. This is an example of

	▼
sentiment analysis.	
speech recognition.	
speech synthesis.	
translation.	

Reference:

<https://azure.microsoft.com/en-gb/services/cognitive-services/speech-to-text/#features>

QUESTION: 97

You need to build an app that will read recipe instructions aloud to support users who have reduced vision. Which version service should you use?

- A. Text Analytics
- B. Translator
- C. Speech
- D. Language Understanding (LUIS)

Answer(s): C

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/text-to-speech/#features>

QUESTION: 98

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
You can use the Speech service to transcribe a call to text.	☐	☐
You can use the Text Analytics service to extract key entities from a call transcript.	☐	☐
You can use the Speech service to translate the audio of a call to a different language.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
You can use the Speech service to transcribe a call to text.	☒	☐
You can use the Text Analytics service to extract key entities from a call transcript.	☒	☐
You can use the Speech service to translate the audio of a call to a different language.	☒	☐

Reference:

<https://docs.microsoft.com/en-gb/azure/cognitive-services/text-analytics/overview>

<https://azure.microsoft.com/en-gb/services/cognitive-services/speech-services/>

QUESTION: 99

Your website has a chatbot to assist customers.

You need to detect when a customer is upset based on what the customer types in the chatbot.

Which type of AI workload should you use?

- A. anomaly detection
- B. computer vision
- C. regression
- D. natural language processing

Answer(s): D

Explanation:

Natural language processing (NLP) is used for tasks such as sentiment analysis, topic detection, language detection, key phrase extraction, and document categorization.

Sentiment Analysis is the process of determining whether a piece of writing is positive, negative or neutral.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 100

You plan to develop a bot that will enable users to query a knowledge base by using natural language processing.

Which two services should you include in the solution? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. QnA Maker
- B. Azure Bot Service
- C. Form Recognizer
- D. Anomaly Detector

Answer(s): A,B

Reference:

<https://docs.microsoft.com/en-us/azure/bot-service/bot-service-overview-introduction?view=azure-bot-service-4.0>

<https://docs.microsoft.com/en-us/azure/cognitive-services/luis/choose-natural-language-processing-service>

QUESTION: 101

In which two scenarios can you use a speech synthesis solution? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. an automated voice that reads back a credit card number entered into a telephone by using a numeric keypad
- B. generating live captions for a news broadcast
- C. extracting key phrases from the audio recording of a meeting
- D. an AI character in a computer game that speaks audibly to a player

Answer(s): A,D

Explanation:

Azure Text to Speech is a Speech service feature that converts text to lifelike speech.

Incorrect Answers:

C: Extracting key phrases is not speech synthesis.

Reference:

<https://azure.microsoft.com/en-in/services/cognitive-services/text-to-speech/>

QUESTION: 102

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
You can use the Translator service to translate text between languages.	☐	☐
You can use the Translator service to detect the language of a given text.	☐	☐
You can use the Translator service to transcribe audible speech into text.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
You can use the Translator service to translate text between languages.	☒	☐
You can use the Translator service to detect the language of a given text.	☒	☐
You can use the Translator service to transcribe audible speech into text.	☐	☒

The translator service provides multi-language support for text translation, transliteration, language detection, and dictionaries.

Speech-to-Text, also known as automatic speech recognition (ASR), is a feature of Speech Services that provides transcription.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/Translator/translator-info-overview>

<https://docs.microsoft.com/en-us/legal/cognitive-services/speech-service/speech-to-text/transparency-note>

QUESTION: 103

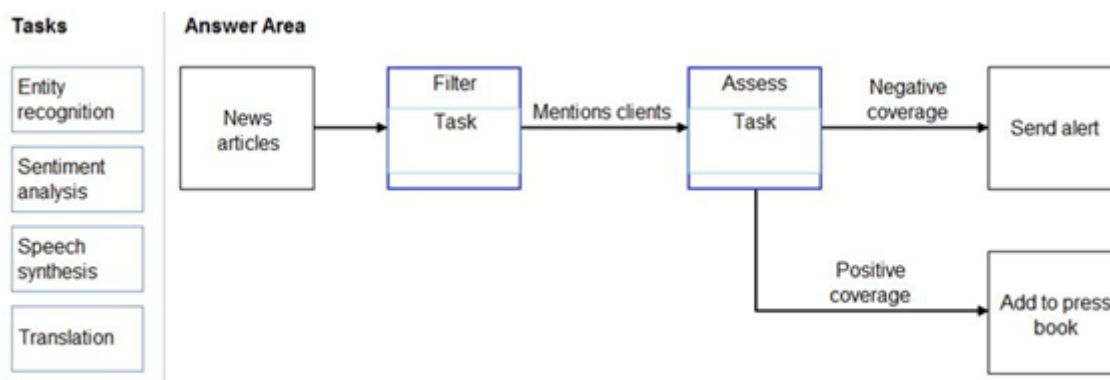
DRAG DROP (Drag and Drop is not supported)

You need to scan the news for articles about your customers and alert employees when there is a negative article. Positive articles must be added to a press book.

Which natural language processing tasks should you use to complete the process? To answer, drag the appropriate tasks to the correct locations. Each task may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

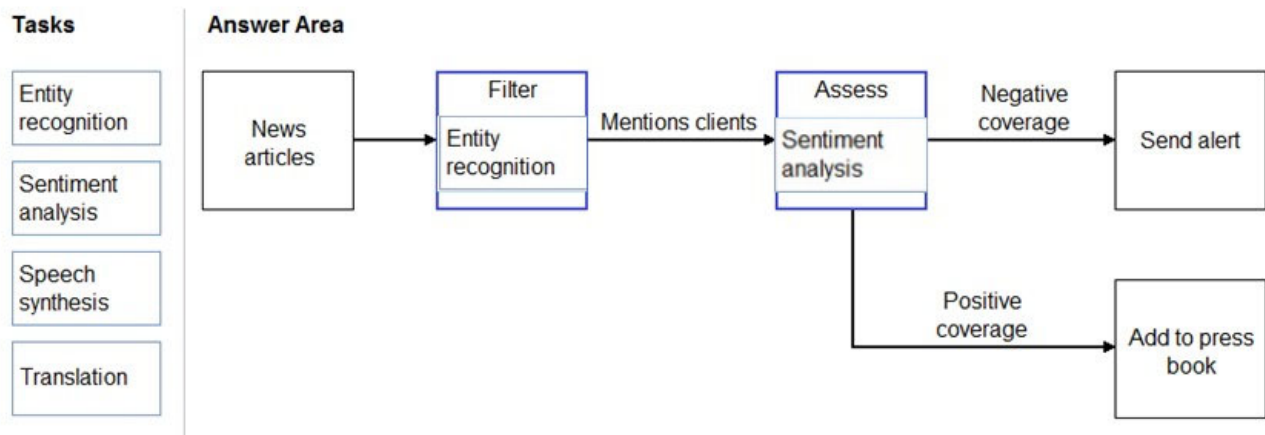
Note: Each correct selection is worth one point.

Select and Place:



Answer(s): A

Explanation:



Box 1: Entity recognition the Named Entity Recognition module in Machine Learning Studio (classic), to identify the names of things, such as people, companies, or locations in a column of text.

Named entity recognition is an important area of research in machine learning and natural language processing (NLP), because it can be used to answer many real-world questions, such as:

- Which companies were mentioned in a news article?
- Does a tweet contain the name of a person? Does the tweet also provide his current location?
- Were specified products mentioned in complaints or reviews?

Box 2: Sentiment Analysis

The Text Analytics API's Sentiment Analysis feature provides two ways for detecting positive and negative sentiment. If you send a Sentiment Analysis request, the API will return sentiment labels (such as "negative", "neutral" and "positive") and confidence scores at the sentence and document-level.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/named-entity-recognition> <https://docs.microsoft.com/en-us/azure/cognitive-services/text-analytics/how-tos/text-analytics-how-to-sentiment-analysis>

QUESTION: 104

You are building a knowledge base by using QnA Maker.

Which file format can you use to populate the knowledge base?

- A. PPTX
- B. XML
- C. ZIP
- D. PDF

Answer(s): D

Explanation:

D: Content types of documents you can add to a knowledge base:

Content types include many standard structured documents such as PDF, DOC, and TXT.

Note: The tool supports the following file formats for ingestion:

-.tsv: QnA contained in the format Question(tab)Answer.

-.txt, .docx, .pdf: QnA contained as regular FAQ content--that is, a sequence of questions and answers.

Incorrect Answers:

A: PPTX is the default presentation file format for new PowerPoint presentations.

B: It is not possible to ingest xml file directly.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/qnamaker/concepts/data-sources-and-content>

QUESTION: 105

In which scenario should you use key phrase extraction?

- A. identifying whether reviews of a restaurant are positive or negative
- B. generating captions for a video based on the audio track
- C. identifying which documents provide information about the same topics
- D. translating a set of documents from English to German

Answer(s): C

QUESTION: 106

You have insurance claim reports that are stored as text.

You need to extract key terms from the reports to generate summaries.

Which type of AI workload should you use?

- A. natural language processing
- B. conversational AI
- C. anomaly detection
- D. computer vision

Answer(s): A

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 107

HOTSPOT (Drag and Drop is not supported)

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Natural language processing can be used to

	▼
classify email messages as work-related or personal.	
predict the number of future car rentals.	
predict which website visitors will make a transaction.	
stop a process in a factory when extremely high temperatures are registered.	

Answer(s): A

Explanation:

Answer Area

Natural language processing can be used to

	▼
classify email messages as work-related or personal.	
predict the number of future car rentals.	
predict which website visitors will make a transaction.	
stop a process in a factory when extremely high temperatures are registered.	

Natural language processing (NLP) is used for tasks such as sentiment analysis, topic detection, language detection, key phrase extraction, and document categorization.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 108

Which AI service can you use to interpret the meaning of a user input such as `Call me back later?`

- A. Translator
- B. Text Analytics
- C. Speech
- D. Language Understanding (LUIS)

Answer(s): D

Explanation:

Language Understanding (LUIS) is a cloud-based AI service, that applies custom machine-learning intelligence to a user's conversational, natural language text to predict overall meaning, and pull out relevant, detailed information.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/luis/what-is-luis>

QUESTION: 109

You are developing a chatbot solution in Azure.

Which service should you use to determine a user's intent?

- A. Translator
- B. QnA Maker
- C. Speech
- D. Language Understanding (LUIS)

Answer(s): D

Explanation:

Language Understanding (LUIS) is a cloud-based API service that applies custom machine-learning intelligence to a user's conversational, natural language text to predict overall meaning, and pull out relevant, detailed information.

Design your LUIS model with categories of user intentions called intents. Each intent needs examples of user utterances. Each utterance can provide data that needs to be extracted with machine-learning entities.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/luis/what-is-luis>

QUESTION: 110

You need to make the written press releases of your company available in a range of languages.

Which service should you use?

- A. Translator
- B. Text Analytics
- C. Speech
- D. Language Understanding (LUIS)

Answer(s): A

Explanation:

Translator is a cloud-based machine translation service you can use to translate text in near real-time through a simple REST API call. The service uses modern neural machine translation technology and offers

statistical machine translation technology. Custom Translator is an extension of Translator, which allows you to build neural translation systems.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/translator/>

QUESTION: 111

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
The Text Analytics service can identify in which language text is written.	☐	☐
The Text Analytics service can detect handwritten signatures in a document.	☐	☐
The Text Analytics service can identify companies and organizations mentioned in a document.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
The Text Analytics service can identify in which language text is written.	☒	☐
The Text Analytics service can detect handwritten signatures in a document.	☐	☒
The Text Analytics service can identify companies and organizations mentioned in a document.	☒	☐

The Text Analytics API is a cloud-based service that provides advanced natural language processing over raw text, and includes four main functions: sentiment analysis, key phrase extraction, named entity recognition, and language detection.

Box 1: Yes

You can detect which language the input text is written in and report a single language code for every

document submitted on the request in a wide range of languages, variants, dialects, and some regional/cultural languages. The language code is paired with a score indicating the strength of the score.

Box 2: No

Box 3: Yes

Named Entity Recognition: Identify and categorize entities in your text as people, places, organizations, date/time, quantities, percentages, currencies, and more.

Well-known entities are also recognized and linked to more information on the web.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/text-analytics/overview>

QUESTION: 112

DRAG DROP (Drag and Drop is not supported)

Match the types of natural languages processing workloads to the appropriate scenarios.

To answer, drag the appropriate workload type from the column on the left to its scenario on the right. Each workload type may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Workloads Types

Entity recognition
Key phrase extraction
Language modeling
Sentiment analysis
Translation
Speech recognition and speech synthesis

Answer Area

Workload Type	Extracts persons, locations, and organizations from the text
Workload Type	Evaluates text along a positive-negative scale
Workload Type	Converts text to a different language

Answer(s): A

Explanation:

Workloads Types

Entity recognition
Key phrase extraction
Language modeling
Sentiment analysis
Translation
Speech recognition and speech synthesis

Answer Area

Entity recognition	Extracts persons, locations, and organizations from the text
Sentiment analysis	Evaluates text along a positive-negative scale
Translation	Converts text to a different language

Box 1: Entity recognition

Named Entity Recognition (NER) is the ability to identify different entities in text and categorize them into pre-defined classes or types such as: person, location, event, product, and organization.

Box 2: Sentiment analysis

Sentiment Analysis is the process of determining whether a piece of writing is positive, negative or neutral.

Box 3: Translation

Using Microsoft's Translator text API

This versatile API from Microsoft can be used for the following:

Translate text from one language to another.

Transliterate text from one script to another.

Detecting language of the input text.

Find alternate translations to specific text.

Determine the sentence length.

Reference:

<https://docs.microsoft.com/en-in/azure/cognitive-services/text-analytics/how-tos/text-analytics-how-to-entity-linking?tabs=version-3-preview> <https://azure.microsoft.com/en-us/services/cognitive-services/text-analytics>

QUESTION: 113

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Monitoring online service reviews for profanities is an example of natural language processing.	☐	☐
Identifying brand logos in an image is an example of natural languages processing.	☐	☐
Monitoring public news sites for negative mentions of a product is an example of natural language processing.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
Monitoring online service reviews for profanities is an example of natural language processing.	☒	☐
Identifying brand logos in an image is an example of natural languages processing.	☐	☒
Monitoring public news sites for negative mentions of a product is an example of natural language processing.	☒	☐

Box 1: Yes

Content Moderator is part of Microsoft Cognitive Services allowing businesses to use machine assisted moderation of text, images, and videos that augment human review.

The text moderation capability now includes a new machine-learning based text classification feature which uses a trained model to identify possible abusive, derogatory or discriminatory language such as slang, abbreviated words, offensive, and intentionally misspelled words for review.

Box 2: No

Azure's Computer Vision service gives you access to advanced algorithms that process images and return information based on the visual features you're interested in. For example, Computer Vision can determine whether an image contains adult content, find specific brands or objects, or find human faces.

Box 3: Yes

Natural language processing (NLP) is used for tasks such as sentiment analysis, topic detection, language detection, key phrase extraction, and document categorization.

Sentiment Analysis is the process of determining whether a piece of writing is positive, negative or neutral.

Reference:

<https://azure.microsoft.com/es-es/blog/machine-assisted-text-classification-on-content-moderator-public-preview/>

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 114

You are developing a natural language processing solution in Azure. The solution will analyze customer reviews and determine how positive or negative each review is.

This is an example of which type of natural language processing workload?

- A. language detection
- B. sentiment analysis
- C. key phrase extraction
- D. entity recognition

Answer(s): B

Explanation:

Sentiment Analysis is the process of determining whether a piece of writing is positive, negative or neutral.

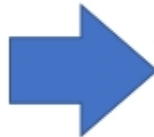
Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 115

You use natural language processing to process text from a Microsoft news story.
You receive the output shown in the following exhibit.

For weeks now, students and teachers have been settling into the uncharted routine of distance learning. Today I want to thank all of the educators who are connecting classrooms and classmates together in the sudden shift to remote learning. This change requires everyone working together and is unlike anything we've seen in the modern history of education. We've seen countries, school districts and universities move rapidly into remote learning environments with Microsoft Teams being used in 175 countries by 183,000 institutions.



now [DateTime]
students [PersonType]
teachers [PersonType]
distance learning [Skill]
Today [DateTime-Date]
educators [PersonType]
classrooms [Location]
classmates [PersonType]
remote learning [Skill]
history [Skill]
education [Skill]
remote learning [Skill]
Microsoft [Organization]
175 [Quantity-Number]
183,000 [Quantity-Number]

Which type of natural languages processing was performed?

- A. entity recognition
- B. key phrase extraction
- C. sentiment analysis
- D. translation

Answer(s): A

Explanation:

Named Entity Recognition (NER) is the ability to identify different entities in text and categorize them into pre-defined classes or types such as: person, location, event, product, and organization.
In this question, the square brackets indicate the entities such as DateTime, PersonType, Skill.

Reference:

<https://docs.microsoft.com/en-in/azure/cognitive-services/text-analytics/how-tos/text-analytics-how-to-entity-linking?tabs=version-3-preview>

QUESTION: 116

DRAG DROP (Drag and Drop is not supported)

You plan to apply Text Analytics API features to a technical support ticketing system.

Match the Text Analytics API features to the appropriate natural language processing scenarios.

To answer, drag the appropriate feature from the column on the left to its scenario on the right. Each feature may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

API Features

Entity recognition

Key phrase extraction

Language detection

Sentiment analysis

Answer Area

API Feature

Understand how upset a customer is based on the text contained in the support ticket.

API Feature

Summarize important information from the support ticket.

API Feature

Extract key dates from the support ticket.

Answer(s): A

Explanation:

API Features

Entity recognition

Key phrase extraction

Language detection

Sentiment analysis

Answer Area

Sentiment analysis

Understand how upset a customer is based on the text contained in the support ticket.

Key phrase extraction

Summarize important information from the support ticket.

Entity recognition

Extract key dates from the support ticket.

Box1: Sentiment analysis

Sentiment Analysis is the process of determining whether a piece of writing is positive, negative or neutral.

Box 2: Broad entity extraction

Broad entity extraction: Identify important concepts in text, including key

Key phrase extraction/ Broad entity extraction: Identify important concepts in text, including key phrases and named entities such as people, places, and organizations.

Box 3: Entity Recognition

Named Entity Recognition: Identify and categorize entities in your text as people, places, organizations, date/time, quantities, percentages, currencies, and more.

Well-known entities are also recognized and linked to more information on the web.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing> <https://azure.microsoft.com/en-us/services/cognitive-services/text-analytics>

QUESTION: 117

You are authoring a Language Understanding (LUIS) application to support a music festival.

You want users to be able to ask questions about scheduled shows, such as: `Which act is playing on the main stage?`

The question `Which act is playing on the main stage?` is an example of which type of element?

- A. an intent
- B. an utterance
- C. a domain
- D. an entity

Answer(s): B

Explanation:

Utterances are input from the user that your app needs to interpret.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/LUIS/luis-concept-utterance>

QUESTION: 118

You build a QnA Maker bot by using a frequently asked questions (FAQ) page.

You need to add professional greetings and other responses to make the bot more user friendly.

What should you do?

- A. Increase the confidence threshold of responses
- B. Enable active learning
- C. Create multi-turn questions
- D. Add chit-chat

Answer(s): D

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/qnamaker/how-to/chit-chat-knowledge-base?tabs=v1>

QUESTION: 119

You need to develop a chatbot for a website. The chatbot must answer users' questions based on the information in the following documents:

- A product troubleshooting guide in a Microsoft Word document
- A frequently asked questions (FAQ) list on a webpage

Which service should you use to process the documents?

- A. Azure Bot Service
- B. Language Understanding
- C. Text Analytics
- D. QnA Maker

Answer(s): D

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/QnAMaker/Overview/overview>

QUESTION: 120

You are building a Language Understanding model for an e-commerce business.

You need to ensure that the model detects when utterances are outside the intended scope of the model.

What should you do?

- A. Test the model by using new utterances
- B. Add utterances to the None intent
- C. Create a prebuilt task entity
- D. Create a new model

Answer(s): B

Explanation:

The None intent is filled with utterances that are outside of your domain.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/LUIS/luis-concept-intent>

QUESTION: 121

Which two scenarios are examples of a natural language processing workload? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. monitoring the temperature of machinery to turn on a fan when the temperature reaches a specific threshold
- B. a smart device in the home that responds to questions such as, "What will the weather be like today?"
- C. a website that uses a knowledge base to interactively respond to users' questions
- D. assembly line machinery that autonomously inserts headlamps into cars

Answer(s): B,C

Explanation:

Natural language processing (NLP) is used for tasks such as sentiment analysis, topic detection, language detection, key phrase extraction, and document categorization.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/natural-language-processing>

QUESTION: 122

You have an AI solution that provides users with the ability to control smart devices by using verbal commands.

Which two types of natural language processing (NLP) workloads does the solution use? Each correct answer presents part of the solution.

Note: Each correct selection is worth one point.

- A. text-to-speech
- B. key phrase extraction
- C. speech-to-text
- D. language modeling
- E. translation

Answer(s): B,C

Explanation:

Key phrase extraction is one of the features offered by Azure Cognitive Service for Language, a collection of machine learning and AI algorithms in the cloud for developing intelligent applications that involve written language. Use key phrase extraction to quickly identify the main concepts in text. For example, in the text "The food was delicious and the staff were wonderful.", key phrase extraction will return the main topics: "food" and "wonderful staff".

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/language-service/key-phrase-extraction/overview>

QUESTION: 123

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area**Statements****Yes****No**

The Language service can identify in which language text is written.

☐☐

The Language service can detect handwritten signatures in a document.

☐☐

The Language service can identify companies and organizations mentioned in a document.

☐☐**Answer(s): A****Explanation:****Answer Area****Statements****Yes****No**

The Language service can identify in which language text is written.

☒☐

The Language service can detect handwritten signatures in a document.

☐☒

The Language service can identify companies and organizations mentioned in a document.

☒☐**Box 1: Yes**

Azure Cognitive Service for Language provides features including:

* Language detection: This pre-configured feature evaluates text, and determines the language it was written in. It returns a language identifier and a score that indicates the strength of the analysis.

Box 2: No

Handwritten detection is part of OCR (Optical Character Recognition).

Box 3: Yes

Azure Cognitive Service for Language provides features including:

* Named Entity Recognition (NER): This pre-configured feature identifies entities in text across several pre-defined categories.

Note: Named entity recognition is a natural language processing technique that can automatically scan entire articles and pull out some fundamental entities in a text and classify them into predefined categories.

Entities may be,

Organizations,

Quantities,

Monetary values,

Percentages, and more.

People's names

Company names

Geographic locations (Both physical and political)

Product names

Dates and times

Amounts of money

Names of events

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/language-service/overview>

QUESTION: 124

DRAG DROP (Drag and Drop is not supported)

You plan to use Azure Cognitive Services to develop a voice controlled personal assistant app.

Match the Azure Cognitive Services to the appropriate tasks.

To answer, drag the appropriate service from the column on the left to its description on the right. Each service may be used once, more than once, or not at all.

Note: Each correct selection is worth one point.

Select and Place:

Services

Speech
Language service
Translator Text

Answer Area

Convert a user's speech to text

Identify a user's intent

Provide a spoken response to the user

Answer(s): A

Explanation:

Services

Speech
Language service
Translator Text

Answer Area

Speech
Language service
Speech

Convert a user's speech to text

Identify a user's intent

Provide a spoken response to the user

Box 1: Speech

The Speech service provides speech-to-text and text-to-speech capabilities with an Azure Speech resource. You can transcribe speech to text with high accuracy, produce natural-sounding text-to-speech voices, translate spoken audio, and use speaker recognition during conversations.

Box 2: Language service

Build applications with conversational language understanding, a Cognitive Service for Language feature that understands natural language to interpret user goals and extracts key information from conversational phrases. Create multilingual, customizable intent classification and entity extraction models for your domain- specific keywords or phrases across 96 languages.

Box 3: Speech

Incorrect:

Not Translator text: Text translation is a cloud-based REST API feature of the Translator service that uses neural machine translation technology to enable quick and accurate source-to-target text translation in real time across all supported languages.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/overview>

<https://azure.microsoft.com/en-us/services/cognitive-services/conversational-language-understanding/>

<https://docs.microsoft.com/en-us/azure/cognitive-services/translator/text-translation-overview>

QUESTION: 125

You need to make the written press releases of your company available in a range of languages.

Which service should you use?

- A. Speech
- B. Language
- C. Translator
- D. Personalizer

Answer(s): C

Explanation:

Translator, an AI service for real-time document and text translation.

Translate text instantly or in batches across more than 100 languages, powered by the latest innovations in machine translation. Support a wide range of use cases, such as translation for call centers, multilingual conversational agents, or in-app communication.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/translator/4>

QUESTION: 126

You have insurance claim reports that are stored as text.

You need to extract key terms from the reports to generate summaries.

Which type of AI workload should you use?

- A. anomaly detection
- B. natural language processing
- C. computer vision
- D. knowledge mining

Answer(s): B

Explanation:

Key phrase extraction is one of the features offered by Azure Cognitive Service for Language, a collection of machine learning and AI algorithms in the cloud for developing intelligent applications that involve written language. Use key phrase extraction to quickly identify the main concepts in text. For example, in the text "The food was delicious and the staff were wonderful.", key phrase extraction will return the main topics: "food" and "wonderful staff".

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/language-service/key-phrase-extraction/overview>

QUESTION: 127

You need to build an app that will read recipe instructions aloud to support users who have reduced vision. Which version service should you use?

- A. Language service
- B. Translator
- C. Speech
- D. Personalizer

Answer(s): C

Explanation:

Speech, a managed service offering industry-leading speech capabilities such as speech-to-text, text-to-speech, speech translation, and speaker recognition.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/speech-services/>

QUESTION: 128

You have a webchat bot that provides responses from a QnA Maker knowledge base. You need to ensure that the bot uses user feedback to improve the relevance of the responses over time. What should you use?

- A. key phrase extraction
- B. sentiment analysis
- C. business logic
- D. active learning

Answer(s): D

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/qnamaker/how-to/improve-knowledge-base>

QUESTION: 129

You are developing a conversational AI solution that will communicate with users through multiple channels including email, Microsoft Teams, and webchat.

Which service should you use?

- A. Text Analytics
- B. Azure Bot Service
- C. Translator
- D. Form Recognizer

Answer(s): B

Reference:

<https://docs.microsoft.com/en-us/azure/bot-service/bot-service-overview-introduction?view=azure-bot-service-4.0>

QUESTION: 130

HOTSPOT (Drag and Drop is not supported)

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Note: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
A bot that responds to queries by internal users is an example of a conversational AI workload.	☐	☐
An application that displays images relating to an entered search term is an example of a conversational AI workload.	☐	☐
A web form used to submit a request to reset a password is an example of a conversational AI workload.	☐	☐

Answer(s): A

Explanation:

Answer Area

Statements	Yes	No
A bot that responds to queries by internal users is an example of a conversational AI workload.	☒	☐
An application that displays images relating to an entered search term is an example of a conversational AI workload.	☐	☒
A web form used to submit a request to reset a password is an example of a conversational AI workload.	☒	☐

Reference:

<https://docs.microsoft.com/en-us/azure/bot-service/bot-service-overview-introduction?view=azure-bot-service-4.0>

QUESTION: 131

You need to provide content for a business chatbot that will help answer simple user queries.

What are three ways to create question and answer text by using QnA Maker? Each correct answer presents a complete solution.

Note: Each correct selection is worth one point.

- A. Generate the questions and answers from an existing webpage.
- B. Use automated machine learning to train a model based on a file that contains the questions.
- C. Manually enter the questions and answers.
- D. Connect the bot to the Cortana channel and ask questions by using Cortana.
- E. Import chat-chat content from a predefined data source.

Answer(s): A,C,E

Explanation:

Automatic extraction

Extract question-answer pairs from semi-structured content, including FAQ pages, support websites, excel files, SharePoint documents, product manuals and policies.

Reference:

<https://docs.microsoft.com/en-us/azure/cognitive-services/qnamaker/concepts/content-types>

QUESTION: 132

You have a frequently asked questions (FAQ) PDF file.

You need to create a conversational support system based on the FAQ.

Which service should you use?

- A. QnA Maker
- B. Text Analytics
- C. Computer Vision
- D. Language Understanding (LUIS)

Answer(s): A

Explanation:

QnA Maker is a cloud-based API service that lets you create a conversational question-and-answer layer over your existing data. Use it to build a knowledge base by extracting questions and answers from your semi-structured content, including FAQs, manuals, and documents.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/qna-maker/>

Gentle Reminder

Please prepare for the recently added Gen AI under the AI-900 certification which is missing in this list. You can find the necessary resources at [Introduction to Generative AI](#).

If you found this repository helpful, please leave a star. Feel free to reach out to me on [LinkedIn](#) or [Twitter](#) if you need any assistance.v